Memory Overload and Asymmetric Under-reaction*

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June 12, 2025

Abstract

We show that stronger memory overload results in a disproportionately greater under-reaction to belief-challenging information compared to belief-confirming information. In an experiment, we keep constant the information received by subjects but vary the difficulty of memorizing the signals. In the treatment condition, the preceding signal disappears as the next signal appears. In the control condition, the preceding signal remains visible. There is stronger memory overload in the treatment than in the control condition. We find that in the treatment condition, subjects under-react when they receive belief-challenging information but not when they receive belief-confirming information. Further analysis supports the mechanism in which it is more cognitively taxing to memorize belief-confirming than belief-challenging signals. However, it does not support the mechanism in which subjects have a fixed amount of cognitive resources to allocate to different signals. Our results caution against information campaigns that ignore the role of cognitive load and attention.

Keywords: memory overload, belief formation, under-reaction
JEL codes: D83, D91

1 Introduction

Thanks to advances in information technology, information is becoming increasingly abundant. Combined with the fact that we are dealing with increasingly complicated tasks and that multi-tasking is more prevalent, cognitive overload has become a prevalent problem we face daily. Different surveys have reported that a significant proportion of employees around the world experience cognitive overload.[1] The increase in cognitive load could impact how individuals utilize information and negatively affect their decision-making. It is thus vital to understand how cognitive

*Funding by the German Research Foundation (DFG) through CRC TR 224 (Project B07) is gratefully acknowledged. We thank Rafael Suchy and Zhi Hao Lim for their excellent service as research assistants. Benson Tsz Kin Leung would like to acknowledge the financial support from the Digital Chair Jean-Jacques Laffont.

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[1] See, for example, https://www.forbes.com/sites/bryanrobinson/2023/03/10/employees-are-overloaded-96-say-workplace-tools-arent-helping-them-keep-up/ and https://cogitocorp.com/blog/the-impact-of-cognitive-overload-at-work/.

overload affects the processing behavior of individuals. This paper examines a specific yet important aspect of cognitive overload, memory overload, and analyzes how it impacts belief updating with belief-confirming and belief-challenging information.

While there exist many other important aspects of cognitive overload, we focus on memory overload for the following reasons. First, memory is a fundamental cognitive process that is relatively easy to manipulate in the lab. Second, recent experimental and empirical evidence suggests that memory plays a significant role in belief formation (Enke et al. (2024), Graeber et al. (2024), and Bordalo et al. (2025)). Third, a growing theoretical literature shows how memory constraints contribute to confirmation bias, even when individuals adopt an optimal information processing strategy (Wilson (2014), Leung (2020) and Jehiel and Steiner (2019)).

Studying the impact of memory overload in an empirical setting is trickier than one would think. First, in typical behavioral experiments, subjects often see one signal at a time and report their belief in between and are likely to be subject to cognitive/memory overload. Second, and more importantly, it is important to rule out confounding variables and disentangle the effects of information and memory overload. One must introduce a stronger memory overload without (or with minimal) alternation in the information that subjects see. Suppose an experimenter varies the level of memory overload simply by providing more information to the treatment than to the control group. As a result, the impact of more information and more memory overload is necessarily convoluted. A slightly more thoughtful approach is to provide a larger quantity of weak information in the treatment group and a smaller quantity of strong information in the control group while maintaining the aggregate informativeness constant. However, this approach cannot rule out the possibility that the different processing behaviors towards strong and weak information drive any behavioral differences between the two groups. Lastly, asking subjects to remember distractors may introduce spurious interference with the memorization of the signals themselves.

In our experiment, we disentangle the impact of memory overload by varying the difficulty of memorization while keeping all information received by the subjects fixed. More specifically, in our experiment, subjects receive a sequence of numbers as signals about an underlying state of the world, which could be state "low" or state "high". The "low" distribution is more likely to generate small numbers as signals, while the "high" distribution is more likely to generate large numbers. Subjects in the treatment and control groups have to navigate through the same sequence of 7 numbers within 30 seconds and then report their beliefs of the underlying distribution. In the control condition, as subjects navigate through the sequence, the preceding numbers remain visible, and they observe multiple numbers at the same time. In contrast, in the treatment condition, the previous signals disappear, so they only see one signal at a time and thus have to memorize the previous signals.

As subjects in the control condition see multiple numbers simultaneously, it is easier to develop an idea about the aggregate information conveyed by the numbers. In contrast, belief updating is more cognitively taxing in the treatment condition as subjects have to memorize previous signals. Therefore, the treatment has stronger memory overload than the control condition, and we indeed find evidence that subjects in the treatment condition make larger updating mistakes.

Our results show that memory overload drives asymmetric under-reaction to belief-challenging versus belief-confirming information, which supports the findings of the aforementioned theoretical literature. Indeed, we find that upon receiving belief-challenging information, subjects in the treatment condition under-update more than those in the control condition. On the other hand, upon receiving a belief-confirming signal, subjects' belief-updating behaviors do not differ significantly across the two conditions. Thus, holding the available signals constant, stronger memory overload (empirically) leads to more biased processing behavior. Moreover, the stronger bias is driven by a stronger under-reaction to belief-challenging information, not the updating behavior with belief-confirming information. As a result, subjects in the treatment condition are also less likely to switch between guessing "high" and "low" than in the control condition, even when they receive strong belief-challenging signals. We also document heterogeneous treatment effects, in which subjects who are more overwhelmed by memory overload exhibit stronger treatment-induced under-reaction to belief-challenging information than to belief-confirming information.

Drawing inspiration from the aforementioned theoretical literature, we propose two underlying mechanisms that support our results. The first mechanism assumes that subjects have a fixed memory capacity to allocate to the sequence of signals. Attention or cognitive resources spent on the signals are thus substitutes. The second mechanism assumes that subjects face a per-signal cost to memorize each signal, and the cost is higher for belief-challenging than belief-confirming signals. Here, attention or cognitive resources spent on each signal are independent. The fixed capacity mechanism predicts that if subjects under-react to the first 6 belief-challenging signals, they must expend their residual capacity on the last belief-challenging signals, thereby reducing the treatment-induced under-reaction. In contrast, the per-signal cost mechanism predicts that treatment-induced under-reaction increases the number of belief-challenging signals that subjects receive. We find evidence supporting the per-signal cost mechanism but not the fixed capacity mechanism. This result informs future theoretical modeling of limited cognition and drives different implications.

Our findings that memory overload makes individuals more prone to confirmation bias suggest that besides preferences, the informational environment also contributes to the bias. This channel leads to policy implications that differ from utility-based

mechanisms' policy implications. Much like the experiment reported here, our results imply that one way to weaken confirmation bias is to make information easier to process. Our results also caution against information campaigns that overlook the role of cognitive load or attention. In particular, an information campaign that strengthens memory overload would disproportionally strengthen under-reaction to belief-challenging information, and therefore backfire. Moreover, a working environment with more severe memory overload decreases decision quality. Lastly, our findings are particularly pertinent to different social issues in the current information age, with the leading example being ideological polarization (Gentzkow and Shapiro (2011), Flaxman et al. (2016)).

Generally speaking, our paper also contributes to the experimental literature on how individuals update their beliefs. Eil and Rao (2011), Ertac (2011), Grossman and Owens (2012) and Möbius et al. (2022) study how subjects' beliefs about their task performances, IQ or beauty scores evolve with information, and find evidence supporting the phenomenon of overconfidence. Liang (2025) shows that subjects substantially discount the signals when the quality of the information source is ambiguous. Similar to this paper, recent experimental literature finds that the information environment, or how information is presented, impacts the prevalence of biases and how individuals update their beliefs. Enke and Zimmermann (2019) finds that a large proportion of their subjects neglect the correlation between signals when forming their beliefs, and reducing complexity reduces correlation neglect. Enke (2020) finds that computational complexity affects whether subjects take into account unobserved signals when they update their beliefs. Gonçalves et al. (2021) finds that the complexity of information retraction induces under-reaction.

The remainder of this paper is organized as follows. The next section presents two simple theoretical models to motivate our hypotheses and the underlying mechanisms of our treatment effect. In Section 3, we outline the experimental design. In Section 4, we present descriptive statistics for our sample and our main variables and hypotheses. In Section 5, we present the results. Section 6 offers some concluding remarks and implications on information campaigns, advertisements, and managerial decisions.

2 Models and Intuitions

To motivate the intuition that memory overload could give rise to confirmation bias, we present two (toy) models that correspond to two different underlying mechanisms.

2.1 A Mechanism of Fixed Memory Capacity

The first model demonstrates that memory overload leads to more severe under-reaction to belief-challenging signals than to belief-confirming signals via a channel

of fixed memory capacity.

The following model follows closely with Leung (2020) . While the example does not perfectly match our experimental design, it encompasses all of the key features of our experimental design and illustrates the theoretical foundation of our hypotheses. Consider a subject who has to guess whether the "high" distribution or "low" distribution was randomly chosen in each round to generate the numbers he observes as signals. If he makes the correct guess, he receives 1 unit of utility; otherwise, he receives 0. His prior belief is denoted by $(p_H, 1-p_H)$ where p_H is the prior probability assigned to the "high" distribution. We assume $p_H > 0.5$.

Before the subject makes a guess, he receives at most two signals, denoted by s_1 and s_2. Each signal is a number drawn from the segment $(0, \infty)$. The "high" distribution is more likely to generate a higher number, while the "low" distribution is more likely to generate a lower number. Formally, denote the (continuous) signal distribution in the high and low states as f_H and f_L, respectively. The signal distributions have increasing likelihood ratio and without loss of generality, we assume that:

$$s = \frac{f_H(s)}{f_L(s)}.$$

We also assume that the two signal distributions are symmetric around 1:

$$f_H(s) = f_L(s^{-1}) \text{ for all } s. \tag{1}$$

We analyze the following setting with memory overload. We assume that the subject forms beliefs by including information from his memory but that he can only memorize one of two numbers he sees. More precisely, he engages in the following sequential game: After seeing the first number s_1, he decides whether to spend time and memorize the number. If he chooses to memorize s_1, he forgoes s_2; on the other hand, if he chooses to ignore s_1, he proceeds to the next signal s_2 and memorizes it. The timeline is illustrated in Figure 1. We analyze the dual-self perfect Bayesian equilibrium, such that when self-1 ignores s_1, self-2 forms a rational belief of s_1 that is consistent in equilibrium. We fix self-2's off-equilibrium path belief (if any) as p_H. For ease of exposition, we assume that the subject memorizes the number if he is indifferent.

Proposition 1. *With memory overload, in any equilibrium, there exist thresholds $\overline{s} > \underline{s}$ such that the subject memorizes s_1 if and only if $s_1 \geq \overline{s}$ or $s_1 \leq \underline{s}$. Moreover, if the signal distributions are not too informative, i.e.,*

$$\frac{\int_0^{\frac{1-p_H}{p_H}} f_L(s)\,ds \int_{\frac{1-p_H}{p_H}}^{\infty} f_H(s)\,ds}{\int_0^{\frac{1-p_H}{p_H}} f_H(s)\,ds \int_{\frac{1-p_H}{p_H}}^{\infty} f_L(s)\,ds} > \left(\frac{1-p_H}{p_H}\right)^2,$$

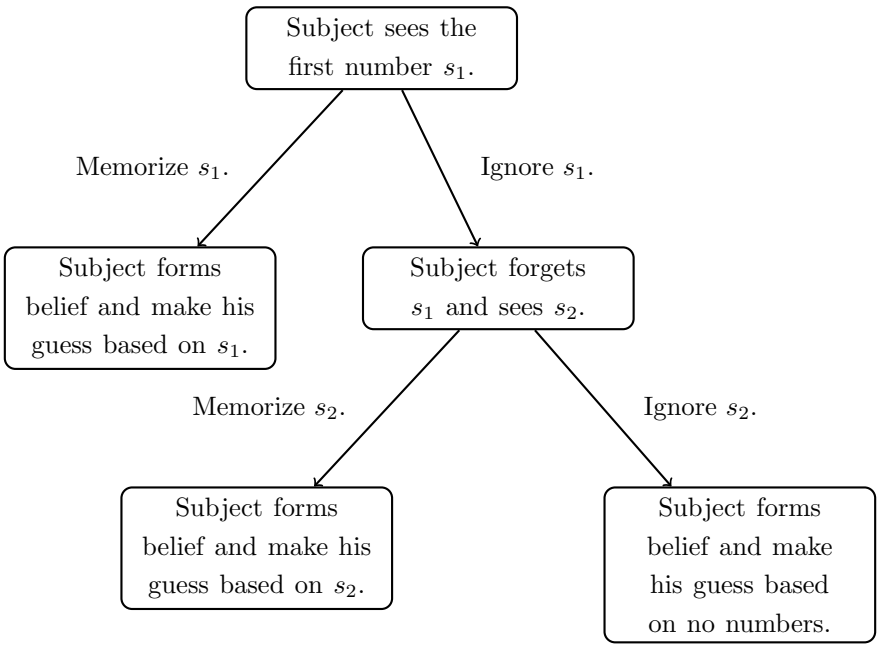


Figure 1: Time-line of the dual-self sequential game.

there always exists a dual-self equilibrium where $\overline{s} < \frac{1}{\underline{s}}$ *such that the subject tends to memorize belief-confirming signals* $s_1 > 0$ *than belief-challenging signals* $s_1 < 0$.

Note that because of the symmetry in signal distributions, when $\overline{s} < \frac{1}{\underline{s}}$, we have 1) the probability of memorizing s_1 in state L is lower than that in state H; 2) the size of belief-challenging signal space $\{s_1 < 0\}$ that the subject memorizes is smaller that that of the belief-confirming signal space $\{s_1 > 0\}$ that the subject memorizes. Hence, when $\overline{s} < \frac{1}{\underline{s}}$, the subject is more likely to ignore belief-challenging signals than belief-confirming signals. The Proposition extends Leung (2020) by finding a sufficient condition that disproportionate under-reaction to belief-challenging signals present all for $p_H > 0.5$.

The proof is presented in Appendix A.1. To understand the intuition of the Proposition, note that the DM trades off between allocating his updating capacity to the current signal s_1 and the future signal s_2. Roughly speaking, he compares the value of the current and future information. When s_1 confirms his belief, he becomes more confident that the "high" distribution was chosen. As a result, the value of the future signal s_2 decreases. In contrast, when s_1 is belief-challenging, the DM's belief moves towards $(\frac{1}{2}, \frac{1}{2})$. This increases the value of the future information s_2 as he becomes more uncertain about the state. This asymmetry in the value of future information drives a symmetric impact of memory overload, as the subject tends to update his belief with a belief-confirming signal and stop looking for future information but ignore a belief-challenging signal and save his cognitive resources for future information. The result suggests that memory overload leads to biased belief-updating behavior. Similar results were also presented in Wilson (2014) and Jehiel and Steiner (2019).

2.2 A mechanism of asymmetric difficulty in memorizing information

Next, we present an alternative mechanism for how memory overload drives asymmetric under-reaction to belief-challenging and belief-confirming information: It is more difficult to memorize belief-challenging information than belief-confirming information.[2] Similar to the previous model, consider a subject who has to guess whether the "high" distribution or "low" distribution was randomly chosen. For simplicity and to distinguish this mechanism better from the mechanism of capacity, assume that the subject only receives one signal $s \in (0, \infty)$ that is drawn from either f_H or f_L. We again assume that $s = \frac{f_H(s)}{f_L(s)}$ and $f_H(s) = f_L(1/s)$. The subject's prior belief is denoted by $(p_H, 1 - p_H)$ where $p_H > 0.5$.

The key assumption in this model is that it is more difficult for the subject to memorize belief-challenging information or, equivalently, more difficult to incorporate belief-challenging information into their belief than belief-confirming information. Formally, we assume that for some $c > 0$, it costs the subject cr^2 to memorize a belief-confirming signal $s \geq 1$ with probability r; it costs the subject δcr^2 with $\delta > 1$ to memorize a belief-challenging signal $s < 1$ with probability r. For simplicity, we assume that once the subject memorizes a signal, he updates his belief to the Bayesian belief p_H^B; otherwise, his belief remains as the prior belief p_H.

Upon receiving a signal s, the subject chooses r to maximizes:

$$-(1-r)\left(\log\frac{p_H^B}{1-p_H^B} - \log\frac{p_H}{1-p_H}\right)^2 - \mathbb{1}_{s\geq 1}cr^2 - \mathbb{1}_{s<1}\delta cr^2.$$

The term $-\left(\log\frac{p_H^B}{1-p_H^B} - \log\frac{p_H}{1-p_H}\right)^2$ captures that the more different the subject's belief is from the Bayesian belief, the subject will incur a higher disutility. The loss could be driven by subsequent decision-making that is based on the subject's "wrong" belief. We assume that the disutility is calculated with the difference between the log-likelihood ratio of the Bayesian and the subject's posterior because of simplicity and to align the model better with the empirical analysis we will perform in later sections.

Denote the optimal probability of memorizing a signal as $r^*(s)$. We have the following result.

Proposition 2. *With memory overload, consider a belief-confirming signal $s > 1$ and a belief-challenging signal $\tilde{s} = 1/s < 1$. The optimal probability of memorizing the belief-challenging signal $\tilde{s}$ is lower than that of memorizing the belief-confirming signal, i.e., $r^*(\tilde{s}) \leq r^*(s)$.*

Moreover, the optimal probability of memorizing the belief-challenging signal $\tilde{s}$ decreases (weakly) relatively more than the optimal probability of memorizing the

[2] See Dellarosa and Bourne Jr (1984) for illustrative evidence.

belief-confirming signal s*, i.e.,* $-\frac{\frac{\partial r^*(\tilde{s})}{\partial c}}{r^*(\tilde{s})} \geq -\frac{\frac{\partial r^*(s)}{\partial c}}{r^*(s)} \geq 0$.

The proof is presented in Appendix A.1. The Proposition shows that for two signals with the same "strength", i.e., s supports state H as much as $\tilde{s}$ supports state L, or mathematically, $\frac{f_H(s)}{f_L(s)} = \frac{f_L(\tilde{s})}{f_H(\tilde{s})}$, a increase in memory load always affects the updating with belief-challenging signals more than with belief-confirming signals. The results are driven by the assumption that memory overload disproportionately affects the memorization of belief-challenging signals more than that of belief-confirming signals. As the two signals have the same "strength", the benefits of memorizing them are equal, but the cost of memorizing the belief-challenging signal is higher.

2.3 Discussion of the two models - Capacity v.s. Per-Signal Cost

The two models both predict qualitatively that stronger memory overload leads to more under-reaction to belief-challenging than to belief-confirming information. However, they also drive different behavioral predictions and implications.

Attention as substitute or independent A key difference between our first and second models is whether memory/attention allocated on different signals are substitute or independent.

In our first model, subjects have a fixed amount of memory capacity to allocate to multiple sequential signals. The asymmetric effect of memory overload is driven by the opportunity cost of forgoing future signals. Thus, when there are no future signals, such an asymmetric effect should be absent. It is optimal to allocate all residual memory capacity to the last signal. Therefore, if the subject ignores a belief-challenging signal in the first stage, he will spend his remaining memory capacity on the second signal even if it is belief-challenging, and therefore reduces any under-reaction (or biases) to the second belief-challenging signal.

In contrast, the second model assumes a mechanism of differential per-signal cost of memorizing belief-confirming and belief-challenging signals. In this model, attention spent on each signal is independent. Unlike the model of fixed capacity, if the subject ignores a belief-challenging signal previously, it does not propel the subject to memorize a subsequent belief-challenging signal and, therefore, does not reduce any under-reaction to the subsequent belief-challenging signal. As a result, biases accumulate as the subject receives more belief-challenging signals.

The different predictions allow us to disentangle the two mechanisms in our analysis.

Different Implications Distinguishing the two mechanisms not only informs the theoretical modeling of belief updating but also has important policy implications.

Both models suggest that policymakers have to consider how information campaigns affect the memory/cognitive load of the target recipients. An information campaign that imposes a strong memory/cognitive load on individuals not only increases the cognitive cost of the individual but also induces a stronger bias that favors the individual's prior belief. This, in turn, reduces the effectiveness of the information campaign.

However, under the first mechanism with fixed memory capacity, a persistent information campaign would eventually force individuals to spend their memory capacity and update their beliefs. No matter how small an individual's memory capacity is, if they receive a sequence of all belief-challenging signals, they will have to memorize these signals, and there will be no ex-post biases. In contrast, under the second mechanism with per-signal cost, even if individuals receive a sequence of all belief-challenging signals, they will under-update with every belief-challenging signal, and the bias remains. If δ is big enough, individuals will update infinmismally with each belief-challenging signal, and thus, a persistent information campaign would remain ineffective. Thus, under the second mechanism with per-signal cost, it is more important to reduce the memory overload imposed by the information campaign.

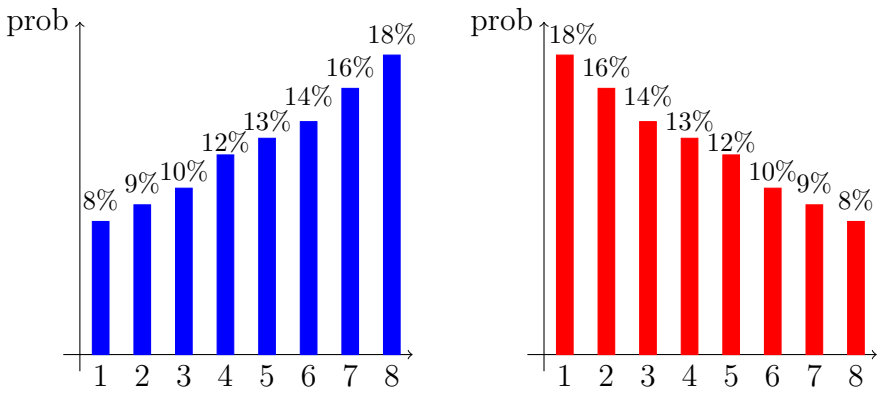
3 Experimental Design

In the experiment, subjects have to complete 12 rounds of a guessing task that involves belief updating with multiple signals.

3.1 States and Information of The Guessing Task

In each round of the guessing task, subjects receive two sequences of numbers drawn independently from either a "high" or "low" distribution. The set of numbers are integers from 1 to 8, inclusive. The probability distribution of a drawn number given a "high" or "low distribution is shown in Figure 2.

As shown in Figure 2, the "high" distribution is more likely to generate larger numbers, while the "low" distribution is more likely to generate smaller numbers. Therefore, subjects can infer which distribution generates the numbers they observe in a round. We explain in detail the reasoning behind the parameters of two distributions in Online Appendix A.2. Briefly speaking, the two distributions are sufficiently informative, so subjects could easily make inferences but not too informative to ensure that the probability of receiving belief-challenging information is significant for the analysis. Moreover, the informativeness of a signal should increase steadily as it goes towards the two extremes (numbers 1 and 8).



(a) The "high" distribution. (b) The "low" distribution.

Figure 2: The two distributions shown in bar charts.

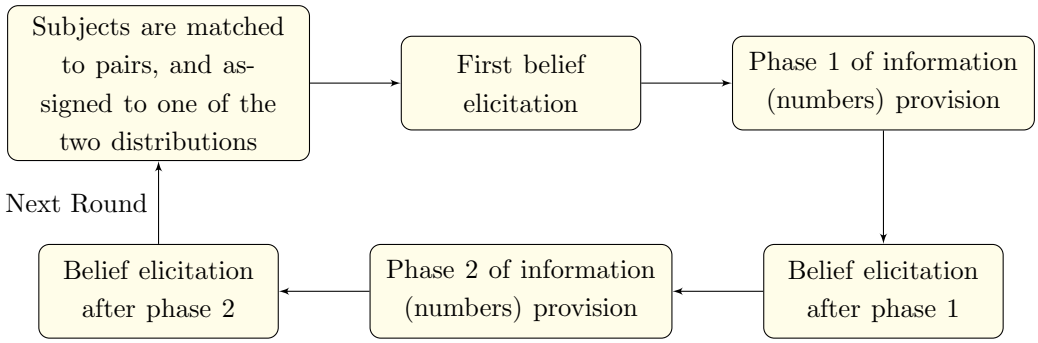


Figure 3: Sequence of a round.

To make the task more accessible to subjects, we call the distributions that generate the numbers "computers" where the "high" ("low") computer is more likely to generate high (low) numbers.

3.2 The procedure of the guessing task

Pairing and assignment of treatment and control. All subjects played 12 rounds of the guessing task and were alternately assigned to the treatment and control conditions. At the beginning of each round, each treatment subject is randomly matched with a control subject to form a pair. During the respective round, the two matched subjects in a pair observe the same sequence of numbers drawn from the same underlying distribution, either the "high" or "low" distribution with equal probability. As mentioned in the introduction, it allows us to single out the effect of memory overload by keeping the two subjects' information constant.

The timeline of a round of the guessing task is illustrated in Figure 3, and we explain in detail below.

First belief elicitation and belief elicitation mechanism. Before the subjects have seen any numbers, we conduct the first belief elicitation at the beginning of

each round. We use a variant of the binarized scoring rule (Hossain and Okui (2013)), which is shown in Figure 4. First, subjects guess whether the "high" or "low" distribution has been selected. Second, they choose between the following two options: earn 8€ if their chosen distribution is selected, or earn 8€ with probability $x\%$, where x starts at 50% and increases in 5% increments per row. The mechanism is incentive-compatible. As an example, if a subject believes the "high" distribution has been chosen with 66%, he should choose "high" for the first question, and for the second question, he should switch from option 1 to option 2 when $x = 70$ as shown in Figure 4.

This first belief elicitation is used to check whether subjects hold the 50-50 belief before seeing any numbers (and that they understand they are at the beginning of a new round). There is a soft time limit of 30 seconds for the belief elicitation.[3] Afterwards, subjects see two sequences of numbers drawn by the selected distribution in two phases, with a second belief elicitation between the two phases and a third belief elicitation after the second phase.

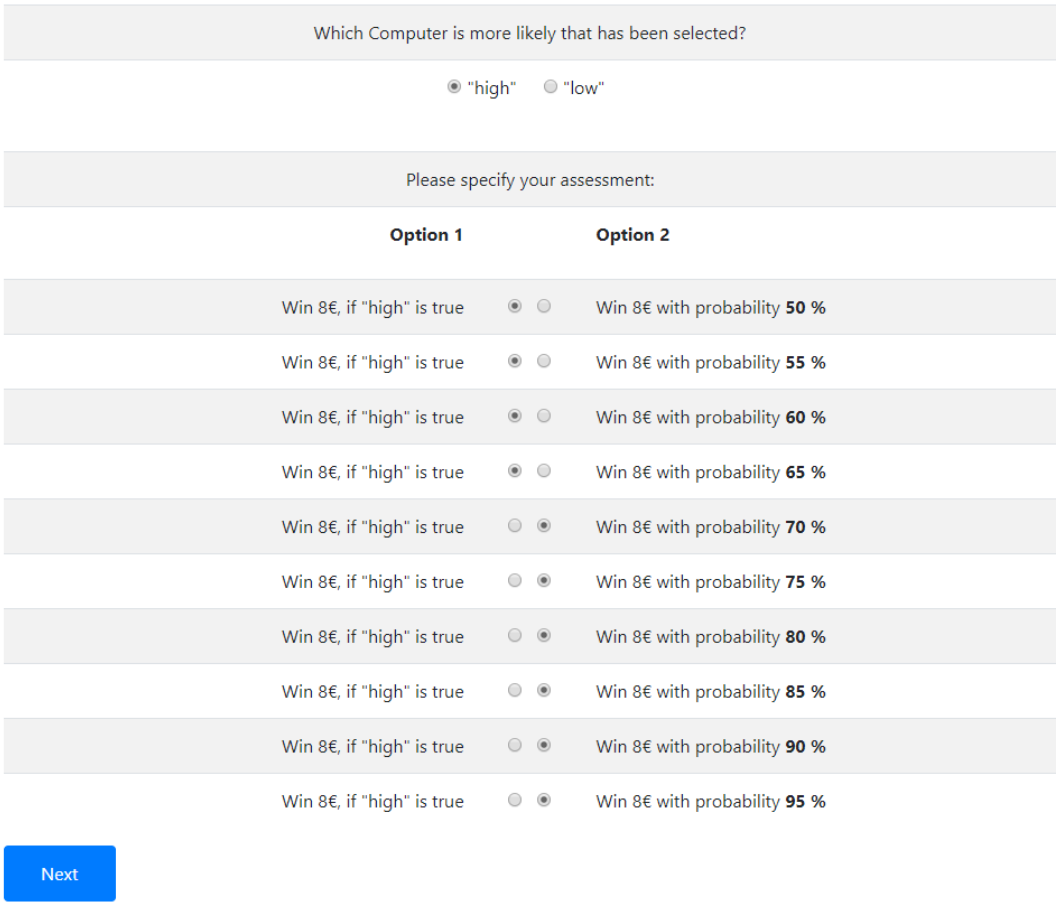


Figure 4: The belief elicitation screen.

[3]If a subject has spent more than 30 seconds, a warning message appears to remind him that time is up. However, subjects are not automatically redirected to the next page.

Phase 1 of information (numbers) provision. In the first phase, all subjects see 5 numbers displayed on the screen for 30 seconds (Figure 5). The two matched subjects from the treatment and control conditions see the same 5 numbers. After 30 seconds, the subjects are redirected to the page of the second belief elicitation.

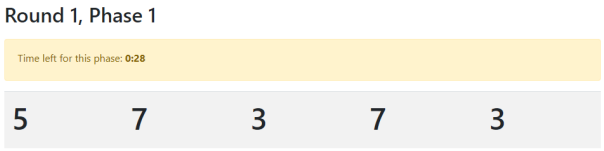


Figure 5: A screenshot of Phase 1.

Belief elicitation after phase 1. After phase 1, we elicit subjects' beliefs using the same table as the first belief elicitation shown in Figure 4. Their choices in the first belief elicitation are shown as a default. The first phase naturally induces heterogeneous beliefs across all the subjects. Thus, it allows us to define belief-confirming and belief-challenging information and studies confirmation bias.

Phase 2 of information (numbers) provision. In the second phase, subjects can see up to 7 numbers with a strict time limit of 30 seconds. Paired subjects in the treatment and control conditions see the same numbers but with a different screen layout.

The treatment condition The layout and flow of the treatment condition are illustrated in Figure 6. The subjects see one number at a time. They can decide when to advance the sequence by clicking the blue "Next" button. Upon clicking "Next", the next number in the sequence is revealed, and the preceding number disappears. Moreover, the subject is unable to return to the preceding numbers. Subjects in the treatment condition face a trade-off between spending more time on the current number and saving time for the next numbers.

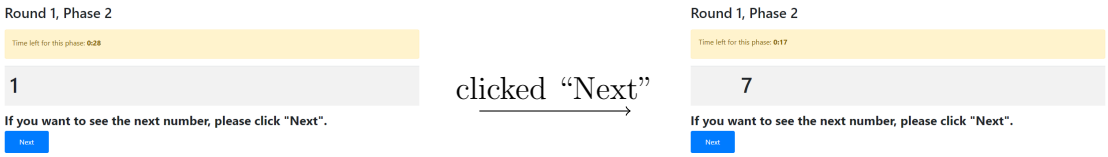


Figure 6: A screenshot of phase 2 in the treatment condition.

The control condition In contrast, subjects in the control condition cannot influence when the next number appears, and the preceding numbers do not disappear when additional numbers are displayed (Figure 7). They start with one number

on their screen, and as they advance through the sequence, they see two, three, four (etc.) numbers on the screen at the same time. To ensure the information they receive and the timing of information provision are the same as their counterpart in the treatment condition, the control subjects advance in the sequence at the same time as their matched treatment subject clicks "Next". The treatment subjects, however, are unaware that they can control the other's advancement in the sequence, nor do the control subjects know that other subjects control their advancement.

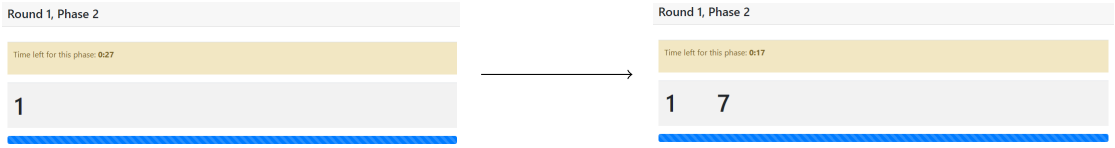


Figure 7: A screenshot of phase 2 in the control condition.

The main difference between the treatment and control conditions is that it is less cognitively taxing in the control than in the treatment conditions. The control subjects do not have to memorize each signals, and it is easier for them to aggregate the information conveyed by the numbers. In other words, subjects in the **control** condition are exposed to **weaker** memory overload than those in the treatment condition, even though they receive the same information. In later sections, we present illustrative evidence that subjects in the treatment condition indeed make more updating mistakes, thus supporting our argument that memory overload is more severe in the treatment than in the control condition.

Belief elicitation after phase 2. After 30 seconds of phase 2, subjects are redirected to the page of the third belief elicitation. Their choices of the second belief elicitation are shown as default. By comparing how subjects update their beliefs in the treatment and control condition with belief-confirming and belief-challenging information, we draw insights into how memory overload gives rise to confirmation bias.

New round. Following the belief elicitation after phase 2, a new round begins where subjects are randomly re-matched and assigned to either the "high" or "low" distribution with equal probability. Subjects are redirected to a screen that reminds them that a new round has started.

3.3 Procedural Details

We conducted 12 sessions of the experiment involving 260 subjects in a university. The participants were university students and recruited through the online recruitment system h-root (Bock et al. (2014)). The experiment was coded and run in

o-Tree (Chen et al. (2016)). Each session took about 2 hours and 15 minutes. The subjects were paid according to a randomly drawn decision in the first, second, and third belief elicitation (Figure 4) from three different randomly chosen rounds. The maximum earning is thus 24€. The average earnings were 18.12€ per subject, plus a participation fee of 7€.

4 Analysis

In this section, we present the descriptive statistics for our sample before constructing and introducing the main variables of interest. After that, we explain our analysis strategy and hypotheses.

4.1 Data

The average age of the subjects was 23, while the maximum age was 30. 109 subjects were male, and 151 subjects were female. 50% of the subjects stated that they had taken economics or statistics courses.

4.1.1 Observations

Each subject in each round provides us with one observation for the guessing task. Each of the 260 subjects played 12 rounds of the guessing task, resulting in a total of 3,120 observations. Unfortunately, there was a technical glitch in the first four sessions with the computer system recording how many numbers a subject saw in phase 2. More specifically, with some probability, the recorded number was one fewer than it should have been, e.g., the computer system may have recorded 6 while the subject has seen 7 numbers in phase 2. Thus, when the recorded number of signals seen in phase 2 is less than 7, it is unclear whether the subjects have not seen all 7 numbers or the glitch has unfortunately affected the observation. Thus, we dropped the observations in the first four sessions if the recorded numbers of signals seen in phase 2 were less than 7, which amounts to 264 of 1008 observations.[4] The technical glitch was fixed in later sessions. After dropping the observations mentioned above, we have 2856 observations. Online Appendix A.4 presents analyses where we exclude all observations in the first 4 sessions, which does not affect our results.

Furthermore, in the analysis, we only use the observations where choices in the first belief elicitation are compatible with the belief that the underlying distribution is "high" or "low" with equal probability, i.e., they chose option 2 in all rows in the

[4] As shown in Table 1 in almost all (97%) of the observations, subjects have seen all 7 numbers in phase 2. Therefore, we are confident that dropping the data does not systematically affect our results. Note that among the 264 of 1008 observations, some subjects have seen all 7 numbers but are not recorded as such. That is why the fraction 264/1008 is much higher than 3% as we observe in sessions 5 to 12.

Table 1: Frequencies and proportion of observations where 4, 5, 6, and 7 numbers have been seen in phase 2.

Number of signals seen in phase 2	All sessions, with only observations where first elicited belief equals $(50\%, 50\%)$		Sessions 5 - 12, with only observations where first elicited belief equals $(50\%, 50\%)$	
	Frequency	Percentage	Frequency	Percentage
4	2	0.09%	2	0.12%
5	9	0.40%	9	0.55%
6	34	1.52%	34	2.07%
7	2196	97.99%	1691	97.27%

Notes. Note that in this table, we do not include observations in sessions 1 to 4 where the number of signals seen in phase 2 is less than 7.

table shown in Figure 4, except possibly in the first row. The first belief elicitation is a robustness check to ensure that subjects understood the belief elicitation mechanism, that they were at the beginning of a new round, and that a new distribution had been drawn with equal probability. In 615 of the 2856 observations (i.e., 21.53%), subjects' choices in the first belief elicitation fail the check.[5,6] Including these 615 observations does not change our results, and we present the analysis in the Online Appendix A.3. After excluding observations in which the first elicited belief is not $(50\%, 50\%)$, we have 2241 final observations. Unless otherwise stated, all the analyses in the main text are based on these observations.

4.1.2 Numbers of Signals Seen in Phase 2

Table 1 shows the descriptive statistics of the number of signals subjects have seen in phase 2. Only a small fraction of subjects have seen less than 7 numbers in phase 2. In sessions 5 to 12, where the technical glitch mentioned above was fixed, fewer than 3% of the subjects saw fewer than 7 numbers. Thus, we are confident that the results presented in this paper are not artificially created by the fact that we have dropped the observations in the first four sessions where the number of signals seen is less than 7.

[5]Among the subjects who have never taken any statistics or economics courses, 26.70% of the observations exhibit choices in the first belief elicitation that are not compatible with the belief of equal probability on the two distributions, while the proportion is 16.27% for the subjects which have taken statistics or economics courses. Throughout the analysis, we control for whether subjects have taken statistics/economics courses.

[6]For example, in 84 observations, subjects' choices in the first belief elicitation indicate that they are at least 95% confident about the underlying distribution (that they chose option 1 in all rows in the table shown in Figure 4). This might be due to a misunderstanding of the belief elicitation mechanism. However, in most cases, subjects realized in later rounds that they were filling out the belief elicitation table incorrectly and did not make the same mistake over all 12 rounds. In fact, only 6 subjects chose option 1 for all rows in the first belief elicitation in more than 6 rounds.

4.1.3 Randomization of High and Low States

Among all 2241 final observations, 48.77% are assigned to the "high" distribution while 51.23% are assigned to the "low" distribution. The composition is not exactly half-half because we have dropped some observations, as mentioned before.

4.1.4 Treatment and Control Condition

Each subject was alternately assigned to the treatment and control conditions in the 12 rounds of the guessing task. Among the final 2241 observations, 1124 observations are from the control condition, and 1117 observations are from the treatment condition. Again, the composition is not exactly half-half because we have dropped some observations, as mentioned before.

4.2 Variables of Interest

4.2.1 Elicited Beliefs

The first elicited belief is denoted by p_0, while the elicited belief after phase 1 and phase 2 are denoted by p_1 and p_2 respectively. All elicited beliefs are normalized such that p_0, p_1 and p_2 represent the (subjective) probability of the "high" distribution being chosen. Note that $p_0 = 0.5$ in all observations used in the analysis, as we exclude those whose first elicited belief does not equal 0.5. On the other hand, p_1 and p_2 are pinned down by the point subjects switch from option 1 to option 2 in the belief elicitation table. As an example, if in the belief elicitation after phase 1, subjects guess "high" for the first question as shown in Figure 4, and switch from option 1 to option 2 when the winning probability of the random lottery is 70%, we define $p_1 = 0.675$, i.e., the average belief compatible with those choices. In the analysis, p_1 is treated as the subjects' prior belief. We investigate how subjects update their beliefs from p_1 to p_2 differently in the control and treatment conditions with belief-confirming or belief-challenging signals.

4.2.2 Bayesian Beliefs

The Bayesian counterpart of the first elicited belief is denoted by p_0^B and correspondingly, the Bayesian counterparts of the elicited beliefs after phase 1 and phase 2 are denoted by p_1^B and p_2^B respectively. Throughout the paper, to ensure comparability with the elicited beliefs, p_1^B and p_2^B are coarsened into bins of size 0.05 and take values equal to the middle of their bins.

p_0^B is always equal to 0.5 as the distribution is drawn with equal probability. The Bayesian belief after phase 1, p_1^B, is constructed using the first elicited belief p_0.

$$\frac{p_1^B}{1-p_1^B} = \prod_{s_i \in \mathscr{S}_1} \frac{f_H(s_i)}{f_L(s_i)} \times \frac{p_0}{1-p_0},$$

where $\mathscr{S}_1$ denotes the set of the 5 numbers in phase 1, and $f_H(s_i)$ is the probability that the "high" distribution generates number s_i and $f_L(s_i)$ is the probability that the number is drawn from the "low" distribution. Since we only include observations where $p_0 = 0.5$, this equals to

$$\frac{p_1^B}{1-p_1^B} = \prod_{s_i \in \mathscr{S}_1} \frac{f_H(s_i)}{f_L(s_i)} \times \underbrace{\frac{0.5}{1-0.5}}_{=1}.$$

Similarly, the Bayesian belief after phase 2, p_2^B, is constructed using the elicited belief after phase 1, p_1:

$$\frac{p_2^B}{1-p_2^B} = \prod_{s_i \in \mathscr{S}_2} \frac{f_H(s_i)}{f_L(s_i)} \times \frac{p_1}{1-p_1},$$

where $\mathscr{S}_2$ is the set of numbers seen in phase 2. In other words, p_2^B is equal to the belief of a Bayesian individual if he takes his prior belief p_1 as given and updates his belief with $\mathscr{S}_2$ in a statistically optimal way.[7]

4.2.3 Treatment and control condition

For each observation i, the condition imposed on the subject is denoted by the dummy variable T_i, which takes on the value of 1 if the subject is assigned to the treatment condition and 0 otherwise.

4.3 Quantifying Bias and Our Main Hypotheses

To examine how memory overload plays a role in belief formation, we analyze the treatment effect on belief changes.

We first quantify subjects' biases in belief formation. We proceed by drawing an analogy between the evolution of the elicited belief and the Bayesian formula.

Consider a subject whose elicited belief after phase 1 is equal to p_1. After he has seen n numbers in phase 2, in which the set is denoted as $\mathscr{S}_2$, his Bayesian belief after phase 2, p_2^B, is given by:

$$\frac{p_2^B}{1-p_2^B} = \prod_{s_i \in \mathscr{S}_2} \frac{f_H(s_i)}{f_L(s_i)} \times \frac{p_1}{1-p_1}.$$

For simplicity of notation, we sometimes denote $\prod_{s_i \in \mathscr{S}_2} \frac{f_H(s_i)}{f_L(s_i)}$ by y_{obj}, or as the "objective odds ratio". Note that the objective odds ratio is a sufficient statistic for

[7] We believe it is unlikely that subjects would revise their belief with respect to the signals in phase 1 during phase 2, given the short time limit and memory overload. Furthermore, the previously elicited belief is always shown as a default in the next belief elicitation, which provides an anchor to distinguish the belief formation in different phases.

a Bayesian individual to update his belief and measures "how much" the subject should update his belief.

We now use the elicited beliefs after phase 1 (p_1) and the elicited beliefs after phase 2 (p_2) to characterize the subjective counterpart of the objective odds ratio, which is denoted as y_{sub}:

$$\begin{aligned} \frac{p_2}{1-p_2} &= y_{sub} \times \frac{p_1}{1-p_1} \\ y_{sub} &= \frac{p_1}{1-p_1} \times \frac{1-p_2}{p_2}. \end{aligned} \tag{2}$$

y_{sub} measures the subject's perceived relative likelihood of seeing the numbers in $\mathscr{S}_2$ with the "high" distribution over that with the "low" distribution. When y_{sub} increases, the subject is more biased towards the "high" distribution; when y_{sub} decreases, the subject is more biased towards the "low" distribution. We denote the subjective odds ratio of the subjects in the treatment and control condition by y_{sub}^T and y_{sub}^C respectively. If subjects are more biased towards distribution they a priori believed in before seeing the numbers in phase 2 in treatment vis-a-vis control setting, we should have the following inequalities:

$$\begin{cases} y_{sub}^T > y_{sub}^C \text{ if } p_1 > 0.5; \\ y_{sub}^C > y_{sub}^T \text{ if } p_1 < 0.5. \end{cases}$$

Equivalently, the two inequalities can be rewritten as

$$-1^{\mathbb{1}_{p_1<0.5}} \left(\frac{y_{sub}^T}{y_{sub}^C} - 1 \right) > 0. \tag{3}$$

4.3.1 Main hypotheses

Our main analysis focuses on how memory overload asymmetrically affects belief updating, particularly with challenging information compared to belief-confirming information, thereby inducing a bias in favor of subjects' prior beliefs. We organize our analysis into a series of 4 hypotheses.

Hypothesis 1. *Memory overload affects subjects' updating bias with belief-confirming and belief-challenging differently. That is, the difference in the ratio $\frac{p_2}{p_2^B}$ in treatment versus control setting, represented by the ratio of subject odds ratio $\frac{y_{sub}^T}{y_{sub}^C}$, are different when subjects receive belief-confirming vis-a-vis belief-challenging information.*

Note that Hypothesis 1 does not test whether the treatment induces biases towards or against subjects' prior belief with each type of information. The following two hypotheses examine the treatment effect on subjects' biases with confirmatory and challenging information separately.

Hypothesis 2. *When subjects receive in aggregate belief-challenging information in phase 2, subjects' posterior belief in the treatment is more biased towards the state which they a priori believe in. Formally, when* $p_1 > 0.5$ *(resp.* $p_1 < 0.5$*),* p_2 *is higher (resp. lower) in the treatment than in the control setting, or equivalently* $y^T_{sub} > y^C_{sub}$ *(resp.* $y^T_{sub} < y^C_{sub}$*).*

Hypothesis 3. *When subjects receive in aggregate belief-confirming information in phase 2, subjects' posterior belief in the treatment is more biased towards the state which they a priori believe in. Formally, when* $p_1 > 0.5$ *(resp.* $p_1 < 0.5$*),* p_2 *is higher (resp. lower) in the treatment than in the control setting, or equivalently* $y^T_{sub} > y^C_{sub}$ *(resp.* $y^T_{sub} < y^C_{sub}$*).*

Note that in Hypothesis 2 and 3, we define belief-confirming and belief-challenging with the aggregate information in phase 2, i.e., whether $y_{obj} > 1$. We will relax this in Section 5.3 where we investigate the underlying mechanisms of the treatment effect.

Hypothesis 2 essentially hypothesizes that subjects under-react more (or over-react less) to belief-challenging information in the treatment condition compared to the control condition. In contrast, Hypothesis 3 essentially hypothesizes that subjects under-react less (or over-react more) to belief-challenging information in the treatment condition compared to the control condition. Both are in favor towards subjects' prior belief.

Now, to establish whether, on average, memory overload (the treatment condition) leads to biases that are in favor of subjects' prior beliefs, the following Hypothesis compares the treatment effect with belief-confirming information and with belief-challenging information and tests whether treatment propels subjects to under-react more to belief-challenging signals than to belief-confirming signals.

Hypothesis 4. *The treatment induced under-reaction with belief challening information, defined as* $\frac{y^T_{sub}}{y^C_{sub}}$*, is larger than the treatment induced under-reaction with belief-confirming information, defined as* $\frac{y^C_{sub}}{y^T_{sub}}$*.*

In a nutshell, Hypothesis 2 and Hypothesis 3 analyze the difference in belief updating with belief-challenging information in the treatment vis-a-vis control condition. Hypothesis 1 and 4 is a difference-in-difference analysis that looks into the difference of the treatment effect with belief-confirming and belief-challenging information. The theoretical analysis in Section 2 suggests that Hypothesis 2 and 4 will hold, while the two models do not support Hypothesis 3.

5 Results

5.1 Preliminaries

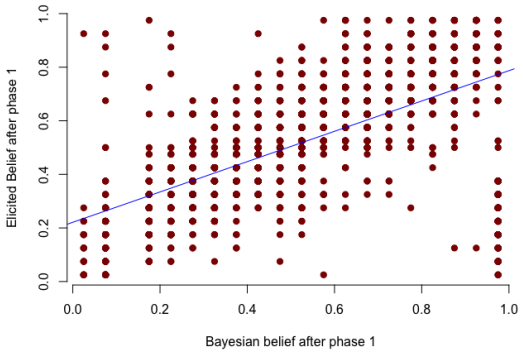
Before we present the main results for the treatment effect of memory overload, we analyze the relationship between elicited belief and Bayesian belief. While it is not the main focus of this paper, the results in this subsection give us a rough idea of how "Bayesian" subjects behave. More importantly, we can ascertain whether subjects understand the experiment well and whether our belief-elicitation mechanism works well in eliciting "normal" behavior of belief-updating, and test whether the treatment condition is indeed more cognitively taxing than the control condition.

Figures 8a, 8b and 8c show the scatter plots and simple regression lines of elicited beliefs against Bayesian beliefs after phase 1 and phase 2 in the treatment and control condition, respectively. From all three figures, we can see a significant and positive correlation between elicited and Bayesian belief, which means that subjects understand the essence of the information structure, i.e., higher numbers serve as stronger evidence that the "high" distribution was chosen in the respective round. On the other hand, as shown in all columns of Table 2 and 3, the regression coefficient of Bayesian belief is smaller than 1. Taken together, both findings suggest that, on average, subjects believe more in the "high" ("low") distribution when they receive higher (lower) numbers, but they tend to under-react to signals compared to the Bayesian benchmark. This result coheres with the findings presented in Eil and Rao (2011) and Liang (2025).

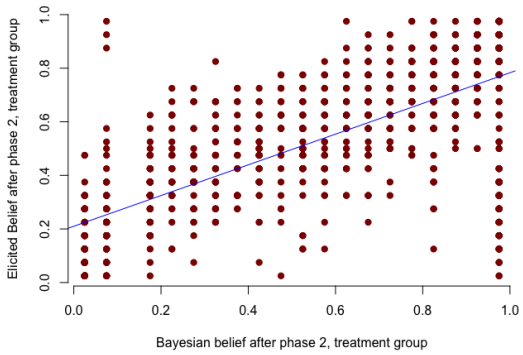
On the other hand, comparing the coefficient of Bayesian belief of the third and foruth column of Table 3, we can see that subjects under-react to both belief-confirming and belief-challenging signals in Phase 2. However, the coefficient is smaller with belief-challenging signals (0.521) than with belief-confirming signals (0.762), suggesting subjects under-react more with belief-challenging signals in phase 2.

Lastly, none of the coefficients on treatment nor the interaction between treatment and Bayesian belief are significant in Table 2 and columns 2 and 3 of Table 3. In contrast, in column 4 of Table 3, where we include only observations where subjects receive a belief-challenging signal in phase 2, the coefficient of treatment is positive and significant, and the coefficient of the interaction between treatment and Bayesian belief is significantly negative. It suggests that subjects in the treatment setting under-react more to belief-challenging signals than subjects in the control setting. The next subsections reaffirm this observation and dive deeper into its underlying mechanism.

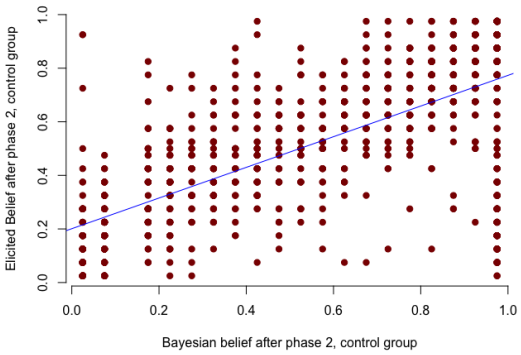
Figure 9a shows the histograms of the absolute difference between elicited and Bayesian belief (the updating mistakes) after phase 1 in the treatment and control conditions, while Figure 9b and Figure 9c show the corresponding histograms



(a) Beliefs after Phase 1.



(b) Beliefs after Phase 2, Treatment.



(c) Beliefs after Phase 2, Control.

Figure 8: Scatter plot and regression line with Bayesian belief on x-axis and Elicited belief on y-axis. Both Bayesian and Elicted beliefs take value of the middle point in their respective 5% bins.

	(1) Elicited belief	(2) Elicited belief
Bayesian Belief after phase 1	0.764***	0.768***
	(0.017)	(0.020)
Treatment		0.001
		(0.013)
Treatment × Bayesian Belief after phase 1		-0.008
		(0.022)
Age Controls	Yes	Yes
Gender Controls	Yes	Yes
Stat/Econ Controls	Yes	Yes
R^2	0.702	0.702
Observations	2241	2241
Subjects	235	235

Clustered standard errors on subject-level in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 2: Regression analysis of elicited beliefs after phase 1.Both Bayesian and Elicted beliefs take value of the middle point in their respective 5% bins. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics courses. Subjects in general underreact to information.

	(1) Elicited belief	(2) Elicited belief	(3) Elicited belief (w confirmatory signals)	(4) Elicited belief (w challenging signals)
Bayesian Belief after phase 2	0.697***	0.708***	0.762***	0.521***
	(0.014)	(0.017)	(0.016)	(0.033)
Treatment		0.017	0.005	0.057*
		(0.010)	(0.011)	(0.024)
Treatment × Bayesian Belief after phase 2		-0.023	-0.002	-0.101*
		(0.015)	(0.014)	(0.041)
Age Controls	Yes	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes	Yes
R^2	0.728	0.729	0.864	0.346
Observations	2241	2241	1327	914
Subjects	235	235	230	225

Clustered standard errors on subject-level in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Regression analysis of elicited beliefs after phase 2. Both Bayesian and Elicted beliefs take value of the middle point in their respective 5% bins. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics courses. Similar to phase 1, subjects in general under-react to information. Moreover, they under-react more to challenging signals than to confirmatory signals.

for beliefs after phase 2 with belief-confirming and challenging signals. The three graphs show that most of the mistakes ($40\% - 60\%$) are less than 10%. Moreover, the frequencies of the mistakes decrease with the magnitude. For example, for belief formation in phase 1 in both treatment and control conditions, almost 60% of the elicited beliefs are within 10% difference of the Bayesian beliefs, while only around 10% of the mistakes are as big as 20%. This again shows that the subjects understand the information structure and are not making big mistakes.

On the other hand, as we can see in Figure 9a, being in the treatment or control condition has no effect on the mistakes made in phase 1, as there are no differences in the settings in phase 1. In particular, the two-sided Kolmogorov–Smirnov test cannot reject the hypothesis that the mistake distribution in the control setting is the same as that in the treatment setting ($p = 0.716$).

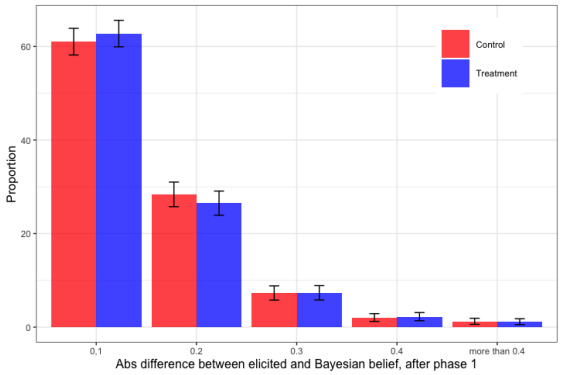
In contrast, we should expect subjects to make more mistakes when updating their belief in phase 2 in the treatment than in the control condition. From Figure 9b, with belief-confirming information, the mistakes in the belief formation in phase 2 in the treatment condition are not significantly different from that in the control condition. The p-value of a two-sided Kolmogorov–Smirnov test is equal to 0.361. In contrast, with belief-challenging information, Figure 9c shows a larger impact of treatment on the mistake distribution, e.g., in the control condition, the share of mistakes smaller than 10% is higher than in the treatment condition (50% of observations) than in control condition (40% of observations), while the share of big mistakes is smaller. However, the differences are not strong enough to reject the null hypothesis in the two-sided Kolmogorov–Smirnov test even though the p-value ($p = 0.189$) is smaller than that for phase 1 beliefs and phase 2 beliefs with confirmatory signals.[8]

5.2 Main Results

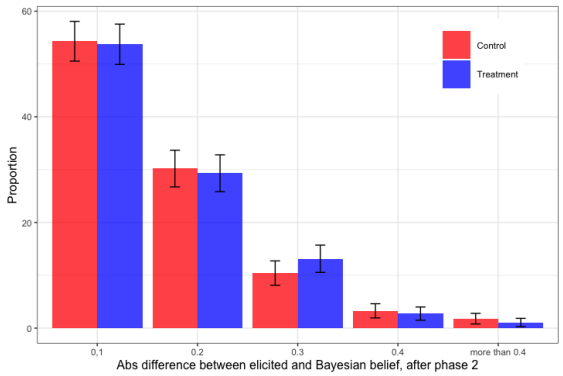
To better illustrate the asymmetric effect of memory overload on belief updating with belief-challenging vis-a-vis belief-confirming information, we now present the regression analysis of the quantified bias. The quantified bias is represented by $\log(y_{sub}) - \log(y_{obj})$. It measures the direction and magnitude of the discrepancy between subjects' belief-updating behavior and the Bayesian benchmark. We will analyze how the treatment condition affects $\log(y_{sub}) - \log(y_{obj})$ in scenarios when subjects receive belief-confirming and belief-challenging information, as discussed in Section 4.3.

We now present the regression specifications for our analysis. The notation is as follows: i denotes the observation. $y_{i,sub}$ and $y_{i,obj}$ denote the subjective and

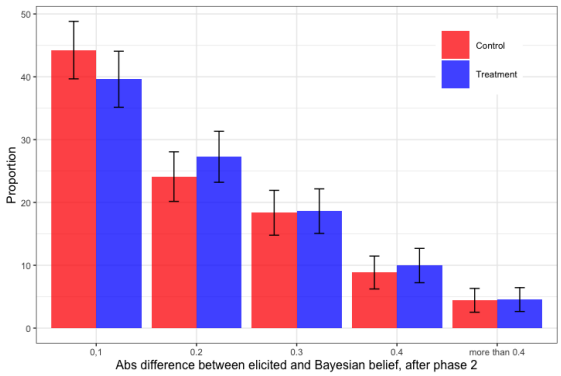
[8]Note that when we include all observations even when the first elicited belief does not equal to 0.5, for updating mistakes after phase 2, a two-sided Kolmogorov–Smirnov test shows that subjects in the treatment condition makes larger mistakes than in the control condition with both belief-challenging and belief-confirming information ($p = 0.04767, 0.06989$ respectively).



(a) Beliefs after Phase 1, treatment versus control.



(b) Beliefs after Phase 2, treatment versus control with confirmatory singals.



(c) Beliefs after Phase 2, treatment versus control with challenging singals.

Figure 9: Distribution of the absolute difference between elicited and Bayesian belief with 95% confidence intervals. Both Bayesian and Elicted beliefs take value of the middle point in their respective 5% bins, and are normalized to values from 0 to 1. Proportions are in percentages.

objective odds ratio of observation i respectively. Challenging_i is a dummy variable which equals to 1 if the phase 2 signals are in aggregate belief-challenging, i.e., $y_{i,obj} < 1$ if $p_1 > 0.5$ and $y_{i,obj} > 1$ if $p_1 < 0.5$. We include fixed effects X_i, which include age, gender, and whether subjects have taken statistics or economics courses. When possible, we will include a pairwise/group fixed effect $\alpha_{m(i)}$ where $m(i)$ denotes the pair that observation i belongs to. In all regression analyses, we cluster standard errors on the subject level.

To test Hypothesis 1, we estimate the following regression:

$$\log(y_{i,sub}) - \log(y_{i,obj}) = \beta_0 + \beta_1 T_i + \beta_2 \text{Challenging}_i + \beta_3 T_i \times \text{Challenging}_i + X_i + \epsilon_i. \quad (4)$$

Our focus is the coefficient of the interaction between the treatment dummy and the challenging signals dummy, where $\text{Challenging}_i = 1$ if $(p_1 - 0.5)y_{i,obj} < 1$. We run the regression with 1) only data with high prior $p_1 > 0.5$; 2) only data with low prior $p_2 < 0.5$; 3) all data where we pool the negative of $\log(y_{i,sub}) - \log(y_{i,obj})$ from low prior observations with $\log(y_{i,sub}) - \log(y_{i,obj})$ from high prior observations. Such pooling leverages Equation (3) and the symmetric nature of the experimental setup, enabling us to improve the statistical power of our analysis.

Table 4 presents the results. First, in column 1, the coefficient β_3 is equal to 0.204 ($p = 0.094$). While the p-value is larger than the 5% significance level, its positive sign suggests treatment induces a larger bias towards the high state than control when subjects receive belief-challenging information than when they receive belief-confirming information. In columns 2 and 3, we can see that the upon receiving belief-challenging information, the treatment effects are significant, and both suggest that memory overload induces a stronger bias in favor of subjects' prior belief when they receive belief-challenging information compared to belief-confirming information. In column 2, the coefficient is significantly negative and equals to -0.254 ($p = 0.045$); in column 3, the coefficient is significantly positive and equals to 0.227 ($p = 0.010$). Both support Hypothesis 1 and further suggest that treatment leads to stronger bias in favor of subjects' prior belief, especially when they receive belief-challenging information.

To test Hypothesis 2 and 3, we separate the data where subjects receive in aggregate belief-challenging and confirming information in phase 2. We run the following regression:

$$\log(y_{i,sub}) - \log(y_{i,obj}) = \beta_0 + \beta_1 T_i + X_i + \alpha_{m(i)} + \epsilon_i, \quad (5)$$

again with high prior, low prior, and all data where we pool the negative of $\log(y_{i,sub}) - \log(y_{i,obj})$ from low prior observations with $\log(y_{i,sub}) - \log(y_{i,obj})$ from high prior observations.

Table 5 provides support to Hypothesis 2, that subjects are more biased to-

	(1) High prior	(2) Low prior	(3) Pooled
Treatment	-0.048	0.093	-0.064
	(0.074)	(0.070)	(0.048)
Challenging	1.149***	-1.068***	1.108***
	(0.103)	(0.100)	(0.079)
Treatment × Challenging	0.209†	-0.235*	0.224**
	(0.124)	(0.116)	(0.086)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Major Controls	Yes	Yes	Yes
R^2	0.273	0.262	0.262
Observations	1219	1022	2241
Subjects	229	222	235

Clustered standard errors on subject-level in parentheses

† $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics courses. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior data with the quantified bias from the high prior data.

wards their prior when they receive belief-challenging information and are subject to stronger memory overload. In other words, they under-react more to belief-challenging signals in the treatment than in the control condition. We focus on the coefficient on the treatment dummy in Table 5. In column 1, the coefficient is equal to 0.183 ($p = 0.057$). While the p-value is slightly larger than the 5% significance level, its positive sign is consistent with Hypothesis 2. Similarly, in column 2, the coefficient is equal to -0.127 ($p = 0.256$). The sign is consistent with Hypothesis 2 although the coefficient is not significant. Lastly, when we pool the low and high prior data, the coefficient is equal to 0.168 and is significantly positive ($p = 0.025$). It suggests that subjects under-react to belief-challenging information in the treatment as

$$\log(y^T_{i,sub}) - \log(y_{i,obj}) - (\log(y^C_{i,sub}) - \log(y^T_{i,sub})) = 0.168,$$

we have $y^T_{i,sub} = \exp(0.168) y^C_{i,sub} = 1.18 \times y^C_{i,sub}$. The subjective odds ratio is 1.18 times higher in the treatment condition than in the control condition.

In contrast, with belief-confirming information, Table 6 does not provide support to Hypothesis 3. All coefficients on the treatment dummy are not significant, i.e., $p = 0.258, 0.330, 0.327$ in columns 1, 2, and 3, respectively. In fact, we cannot reject that subjects under-react more nor less to confirmatory signals in the treatment setting than in the control setting ($p = 0.734$ and $= 0.266$, respectively, when we pool the low and high prior observations).

Lastly, to test whether memory overload induces a stronger under-reaction to belief-challenging information than to belief-confirming (Hypothesis 4), we run a one-tailed t-test to compare the treatment coefficients in Table 5 and the negative of the treatment coefficients in Table 6 In columns 1 and 3, we test whether the treatment coefficient with belief-challenging information is larger (more positive) than with belief-confirming information In column 2, we test whether the treatment coefficient with belief-challenging information is lower (more negative) than with belief-confirming information The one-tailed t-test p-values are equal to 0.232 with high prior observations, 0.362 with low prior observations, and 0.106 when we pool the high and low prior observations In combination, the signs of the coefficients are consistent with Hypothesis 4 although the p-values are higher than the 5% significance level.

5.3 The Mechanism: Fixed Capacity versus Per-Signal Cost

This subsection looks into the underlying mechanisms of the asymmetric effect of memory overload on belief-confirming v.s belief-challenging information. We focus on the two mechanisms discussed in Section 2: 1) subjects have a fixed memory capacity to allocate across different signals; 2) subjects face an independent memory

	(1) High prior w. challenging signals	(2) Low prior w. challenging signals	(3) Pooled w. challenging signals
Treatment	0.183†	-0.127	0.168*
	(0.094)	(0.111)	(0.074)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
Pair Controls	Yes	Yes	Yes
R^2	0.796	0.827	0.803
Observations	516	398	914
Subjects	205	188	225

Clustered standard errors on subject-level in parentheses

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 5: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ when subjects receive in aggregate belief-confirming signals in phase 2, i.e., $(p_1 - 0.5) y_{i,obj} > 0$. Standard errors are clustered on subject-level, and we control for subjects' age, gender, whether they have taken Statistics/Economics courses, and include a pairwise fixed effect. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases.

	(1) High prior w. confirmatory signals	(2) Low prior w. confirmatory signals	(3) Pooled w. confirmatory signals
Treatment	-0.090	0.079	-0.053
	(0.085)	(0.081)	(0.054)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
Pair Controls	Yes	Yes	Yes
R^2	0.794	0.798	0.788
Observations	703	624	1327
Subjects	214	211	230

Clustered standard errors on subject-level in parentheses

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 6: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ when subjects receive in aggregate belief-challenging signals in phase 2, i.e., $(p_1 - 0.5) y_{i,obj} < 0$. Standard errors are clustered on subject-level, and we control for subjects' age, gender, whether they have taken Statistics/Economics courses, and include a pairwise fixed effect. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases.

cost when they memorize each signal.

With the capacity mechanism, as discussed in Section 2, the fixed capacity implies that attention spent on different signals are substitute. Cognitive resources that are not spent on signals 1 to 6 would be spent on signal 7, as there are no future signals beyond the 7th signal. Now, suppose signals 1 to 6 are, in the aggregate, belief-challenging. As a result, when memory overload drives subjects to allocate less memory capacity to signals 1 to 6, they must allocate their residual capacity to signal 7, which effectively reduces the treatment effect on signal 7. We test this prediction by extending the regression Equation (4):

$$\begin{aligned}\log(y_{i,sub}) - \log(y_{i,obj}) =& \beta_0 + \beta_1 T_i + \beta_2 \text{Challenging } 1^{\text{st}}\text{-}6^{\text{th}}{}_i \\ &+ \beta_3 \text{Challenging } 7^{\text{th}}{}_i + \beta_4 \text{Challenging } 1^{\text{st}}\text{-}6^{\text{th}}{}_i \times \text{Challenging } 7^{\text{th}}{}_i \\ &+ \beta_5 T_i \times \text{Challenging } 1^{\text{st}}\text{-}6^{\text{th}}{}_i + \beta_6 T_i \times \text{Challenging } 7^{\text{th}}{}_i \\ &+ \beta_7 T_i \times \text{Challenging } 1^{\text{st}}\text{-}6^{\text{th}}{}_i \times \text{Challenging } 7^{\text{th}}{}_i + X_i + \epsilon_i. \end{aligned} \tag{6}$$

"Challenging 1st-6th" is a dummy variable that equals to 1 if the 6 signals are in aggregate belief-challenging; "Challenging 7th" is a dummy variable that equals to 1 if the 7th signal are in aggregate belief-challenging.

Table 7 presents the results. We focus on the coefficient of the interaction effect between the treatment dummy, the challenging dummy for signals 1 to 6, and the challenging dummy for signal 7, i.e., β_7 in Equation (6). As discussed above, the capacity mechanism predicts a negative β_7 with high prior data, and when we pool low and high prior data, the capacity mechanism predicts a positive β_7 with low prior data. Put differently, if signals 1 to 6 are belief-challenging, residual attention has to be spent on signal 7, which reduces the under-reaction bias, such that $\log(y_{i,sub}) - \log(y_{i,obj})$ decreases if $p_1 > 0.5$ and increases if $p_1 < 0.5$. However, as Table 7 shows, the signs of the coefficient β_7 are not significant in all three columns ($p = 0.393, 0.849, 0.428$ respectively), and their signs are opposite of what the capacity mechanism predicts. Moreover, the treatment effects on the 7th signal is not significantly different from that on the 1st to 6th signals.

Now, with the per-signal cost mechanism, attention/belief updating with the 7 signals are independent. As discussed in Section 2, the treatment effect should thus increase in the number of belief-challenging signals received in phase 2. We present the result in Table 8, with the following regression specification:

$$\log(y_{i,sub}) - \log(y_{i,obj}) = \beta_0 + \beta_1 T_i + \beta_2 \#\text{ Challenging}_i + \beta_3 T_i \times \#\text{ Challenging}_i + X_i + \epsilon_i, \tag{7}$$

where $\#\text{ Challenging}_i$ is the number of challenging signals the subject receives in phase 2. Similar to previous analysis, we run the regression with 1) only high prior data, 2) only low prior data 3) all data where we pool the negative of $\log(y_{i,sub}) -$

	(1) High prior	(2) Low prior	(3) Pooled
Treatment	-0.076	0.025	-0.047
	(0.099)	(0.092)	(0.067)
Challenging 1st-6th	1.312***	-0.976***	1.167***
	(0.121)	(0.109)	(0.090)
Challenging 7th	0.383**	-0.290*	0.342***
	(0.116)	(0.125)	(0.087)
Challenging 1st-6th × Challenging 7th	-0.380†	-0.207	-0.129
	(0.198)	(0.176)	(0.132)
Treatment × Challenging 1st-6th	0.106	-0.008	0.061
	(0.157)	(0.170)	(0.117)
Treatment × Challenging 7th	0.122	-0.040	0.080
	(0.155)	(0.157)	(0.113)
Treatment × Challenging 1st-6th × Challenging 7th	0.199	-0.049	0.132
	(0.232)	(0.258)	(0.166)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.293	0.271	0.273
Observations	1193	1003	2196
Subjects	228	222	235

Clustered standard errors on subject-level in parentheses

† $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 7: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ that test the prediction of the capacity mechanisms. Standard errors are clustered on subject-level, and we control for subjects' age, gender, whether they have taken Statistics/Economics courses. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. "Challenging 1st-6th" is a dummy variable that equals to 1 if the 6 signals are in aggregate belief-challenging. "Challenging 7th" is a dummy variable that equals to 1 if the 7th signal are in aggregate belief-challenging.

$\log(y_{i,obj})$ from low prior observations with $\log(y_{i,sub}) - \log(y_{i,obj})$ from high prior observations. The per-signal mechanism predicts a positive β_3 in 1) and 3) and a negative β_3 in 2).

Table 8 presents the results. In column 1, where we focus on observations with high priors, the coefficient of the interaction between the treatment dummy and the number of phase 2 challenging signals is positive and equal to 0.075 ($p = 0.088$). The sign of the coefficient is thus consistent with the per-signal cost mechanism, albeit its p-value is marginally above the 5% significance level. In column 2, where we focus on observations with low priors, the coefficient of the interaction between the treatment dummy and the number of phase 2 challenging signals is negative and equal to -0.065 ($p = 0.148$). Again, the sign is consistent with the per-signal cost mechanism, but its p-value is higher than the 5% threshold. Lastly, in column 3, we pool low priors data with high priors data to improve the statistical power. The coefficient of the interaction between the treatment dummy and the number of phase 2 challenging signals is positive, equal to 0.068, and is significant ($p = 0.038$). Moreover, comparing the coefficient of the number of phase 2 challenging signals and the coefficient of its interaction with the treatment dummy, the bias induced by memory overload when there are 7 belief-challenging signals is more severe than the bias induced by seeing an additional belief-challenging signal in the control setting, i.e., $0.068 * 7 = 0.476 > 0.370$.

In summary, Tables 7 and 8 together support the mechanism where memory overload drives biases against belief-challenging information through a differential per-signal cost of incorporating belief-challenging signals versus belief-confirming signals, but do not support the mechanism with fixed cognitive capacity.

5.4 Heterogenous Treatment Effect

Before we conclude, we present two additional analyses, i.e., heterogeneous treatment effects across subjects and subjects' switching decisions.

We first examine how the treatment effect varies across individuals. More specifically, we hypothesize that the treatment effect is larger for subjects who are more overwhelmed by our treatment, i.e., those who make larger updating mistakes (the absolute difference between the elicited and Bayesian beliefs) in the treatment condition than in the control condition. To investigate the hypothesis, we defined a variable "subject-level mistakes (treatment-control)" as the average mistakes (the absolute difference between elicited and Bayesian belief after phase 2) that subjects make in their six treatment rounds minus that in their six control rounds: the higher it is, the more overwhelmed they are by our treatment.

We extend the regression Equation (5) by including the subject-level mistakes

	(1) High prior	(2) Low prior	(3) Pooled
Treatment	-0.205	0.189	-0.187
	(0.147)	(0.146)	(0.103)
# Challenging	0.392***	-0.337***	0.370***
	(0.034)	(0.039)	(0.029)
Treatment ×	0.075†	-0.065	0.068*
# Challenging	(0.044)	(0.045)	(0.033)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.295	0.240	0.265
Observations	1193	1003	2196
Subjects	228	222	235

Clustered standard errors on subject-level in parentheses

† $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 8: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ that analyze how the number of challenging signals affects the treatment effect. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics Courses. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. # Challenging$_i$ is the number of challenging signals the subject receives in phase 2.

variable and its interaction with other independent variables:

$$\begin{aligned}\log(y_{i,sub}) - \log(y_{i,obj}) =& \beta_0 + \beta_1 T_i + \beta_2 \text{Subject-level Mistakes}_i \\ &+ \beta_3 T_i \times \text{Subject-level Mistakes}_i + X_i + \epsilon_i.\end{aligned} \tag{8}$$

Similar to Equation (5), we run separate regression analyses for which the subjects receive aggregate belief-challenging and belief-confirming information in phase 2. We focus on the interaction term between the treatment dummy and the subject-level differences in average mistakes in treatment minus control. For example, consider $p_1 > 0.5$, and the subject receives aggregate belief-challenging signals in phase 2. Suppose the coefficient β_3 is positive. In that case, it implies that the more subjects are overwhelmed by our treatment, the more they under-react to belief-challenging signals in the treatment than in the control condition. We again run the regressions with 1) only high prior data, 2) only low prior data, and 3) all data where we pool the negative of $\log(y_{i,sub}) - \log(y_{i,obj})$ from low prior observations with $\log(y_{i,sub}) - \log(y_{i,obj})$ from high prior observations.

First, Table 9 presents the regression analysis where subjects receive in aggregate belief-challenging information in phase 2. In general, the results support that the more overwhelmed subjects are by the stronger memory overload, the more they under-react to belief-challenging signals in the treatment than in the control condition. Except in column 1, where we focus on high prior data only, the coefficients of the interaction between treatment and subject-level mistakes are significant, and the signs are consistent with our hypothesis ($p = 0.506, 0.001, 0.020$ in columns 1, 2, and 3, respectively).

Second, Table 10 presents the regression analysis where subjects receive in aggregate belief-confirming information in phase 2. The results also show that the more overwhelmed subjects are by the stronger memory overload, the more they under-react to belief-confirming signals in the treatment than in the control condition. Except in column 1, where we focus on high prior data only, the coefficients of the interaction between treatment and subject-level mistakes are significant ($p = 0.578, 0.0007, 0.001$ in columns 1, 2, and 3, respectively).

Combining the results in Table 9 and 10, subjects who are more susceptible to memory overload make bigger mistakes not only when they receive belief-challenging information but also when they receive belief-confirming information. However, when comparing the coefficient β_3 with belief-challenging signals and the negative of β_3 with belief-confirming signals, the magnitude of under-reaction induced by memory overload (treatment) and subject-level mistakes is larger with belief-challenging signals than with belief-confirming signals (e.g., $3.611 > -(-2.636)$ in column 3), albeit the difference is not significant ($p = 0.527$ in a two-sided t-test). Moreover, when the subjects receive belief-challenging information, the treatment as a greater relative impact on how individual mistakes affect under-reaction, i.e., the coefficients

of individual mistakes in Table 9 are all insignificant and small compared to the coefficients of the interaction between the treatment dummy and individual mistakes. The same does not hold true when subjects receive belief-challenging information.

	(1) High prior w. challenging signals	(2) Low prior w. challenging signals	(3) Pooled w. challenging signals
Treatment	0.133	-0.104	0.119†
	(0.087)	(0.087)	(0.067)
Subject-level Mistakes	-0.313	0.431	-0.389
(Treatment - Control)	(2.848)	(1.549)	(2.036)
Treatment ×	1.950	-5.399**	3.611*
Subject-level Mistakes	(2.910)	(1.512)	(1.542)
(Treatment - Control)			
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.037	0.102	0.050
Observations	516	398	914
Subjects	205	188	225

Clustered standard errors on subject-level in parentheses

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 9: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ on the heterogeneous treatment effect across subjects when they receive in aggregate belief-challenging signals in phase 2. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics Courses. "Subject-level mistakes (treatment-control)" is defined as the average mistakes (the absolute difference between elicited and Bayesian belief after phase 2) that subjects make in their six treatment rounds minus that in their six control rounds. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases.

5.5 Switching Behavior

Our last set of results relates to how memory overload affects the switching decisions of the subjects. A switch is defined as the scenario where a subject guessed "high" after phase 1 but guessed "low" after phase 2, or vice versa. Moreover, we say that a subject has made a switching mistake when his switching decision differs from that of a Bayesian individual. We analyze two different switching mistakes. The first

	(1) High prior w. confirmatory signals	(2) Low prior w. confirmatory signals	(3) Pooled w. confirmatory signals
Treatment	-0.043	0.057	-0.037
	(0.080)	(0.071)	(0.050)
Subject-level Mistakes	0.956	-4.768***	2.831**
(Treatment - Control)	(1.516)	(1.594)	(1.183)
Treatment ×	-0.704	4.459***	-2.636
Subject-level Mistakes	(2.142)	(1.871)	(1.350)
(Treatment - Control)			
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.034	0.117	0.041
Observations	703	624	1327
Subjects	214	211	230

Clustered standard errors on subject-level in parentheses

† $p < 0.10$,* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 10: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ on the heterogeneous treatment effect across subjects when they receive in aggregate belief-confirming signals in phase 2. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics Courses. We did not include pairwise fixed effects as they introduce collinearity issues with the subject-level mistakes. "Subject-level mistakes (treatment-control)" is defined as the average mistakes (the absolute difference between elicited and Bayesian belief after phase 2) that subjects make in their six treatment rounds minus that in their six control rounds. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases.

mistake is where the subjects should have switched if they were Bayesian but ended up not switching; the second mistake is where the subjects should not have switched if they were Bayesian but ended up switching. We have the following hypotheses:

Hypothesis 5. *Subjects in the treatment condition are strictly less likely to switch decisions when they should, compared to their counterparts in the control condition.*

Hypothesis 6. *Subjects in the treatment condition are strictly less likely to switch decisions when they should not, compared to their counterparts in the control condition.*

It is worth noting that a subject should switch when he receives belief-challenging signals of sufficient strength, while he should not switch when he receives belief-confirming or weak belief-challenging signals. Thus, Hypothesis 5 and Hypothesis 6 correspond to different scenarios: the former examines the subjects' belief-updating behavior with strong belief-challenging signals while the latter analyzes the subjects' belief updating behavior with belief-confirming or weak belief-challenging signals.

Recall that our main results show that subjects in the treatment condition under-react more to belief-challenging information than in the control condition, while their belief-updating behaviors towards belief-confirming information are not significant across the treatment and control condition. As a result, we expect that Hypotheses 5 should hold while 6 may not hold.

We now present the regression specifications for our analysis. We denote $\text{Switch}_i = 1$ if the subject switched decisions in observation i, and $\text{Switch}_i = 0$ otherwise. For Hypothesis 5, we estimate the following regression for observations i where a theoretical Bayesian subject should switch, i.e., where $(p_2^B - 0.5)(p_1 - 0.5) < 0$:

$$1 - \text{Switch}_i = \beta_0 + \beta_1 T_i + X_i + \alpha_{m(i)} + \epsilon_i. \tag{9}$$

β_1 measures the treatment effect on switching mistake (not switching when the subject should switch), and Hypothesis 5 translates to $\beta_1 > 0$. Similarly, for Hypothesis 6, we estimate the following regression for observations i where a theoretical Bayesian subject should not switch:

$$\text{Switch}_i = \beta_0 + \beta_1 T_i + X_i + \alpha_{m(i)} + \epsilon_i, \tag{10}$$

and similarly Hypothesis 6 translates to $\beta_1 < 0$.

Table 11 presents the results. As shown in column 1, the treatment effect is positive and significant ($p = 0.037$) in the scenario where the subject should have switched but did not, which confirms Hypothesis 5. When there is strong enough belief-challenging information so the subjects should switch, subjects in the treatment condition are 10.4% less likely to do so than subjects in the control condition. This is due to the fact that they under-react to belief-challenging signals.

	(1) Should switch but didn't	(2) Shouldn't switch but did
Treatment	0.104*	-0.006
	(0.049)	(0.012)
Age Controls	Yes	Yes
Gender Controls	Yes	Yes
Stat/Econ Controls	Yes	Yes
Group Controls	Yes	Yes
R^2	0.850	0.749
Observations	592	1649
Subjects	207	233

Clustered standard errors on subject-level in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 11: Regression analysis of Switching Decisions after phase 2. Standard errors are clustered on subject-level, and we control for subjects' age, gender, whether they have taken Statistics/Economics courses, and include pairwise fixed effects. The first column focuses on cases where a Bayesian should switch from guessing high in phase 1 to low in phase 2 or guessing low in phase 1 to high in phase 2, but the subject did not switch. The second column focuses on cases where a Bayesian should not switch, but the subject has switched from guessing high to low or guessing low to high. The dependent variables are the dummy variables that denote such switching mistakes.

On the other hand, in the scenario where the subject should not switch but ended up switching, the treatment effect is not significant ($p = 0.642$), which does not confirm Hypothesis 6. This result is consistent with our main results presented in Table 6, where the treatment effect on updating with confirmatory signals is not significantly different from 0. When there is no strong enough belief-challenging information such that subjects should stick to their prior belief, the magnitude of memory overload has no significant effect on switching behavior.

Combining the two findings, we can see that memory overload asymmetrically affects individuals' switching decisions when they receive different types of information. More specifically, the effect is significant only when the subjects should have switched, i.e., they receive strong enough belief-challenging information. Roughly speaking, subjects react less to belief-challenging information when exposed to stronger memory overload. In contrast, their reaction to belief-confirming information (that does not induce switching) is not significantly affected by memory overload.

The asymmetric effect also speaks against another possible hypothesis that subjects switch less under stronger memory overload only because they under-react to every (belief-challenging and belief-confirming) signal they receive instead of being more biased. Suppose, in contrast, the subjects under-react more to both belief-confirming and belief-challenging information in the treatment condition; they then will be more reluctant to update their beliefs towards the extreme when they receive belief-confirming information. It implies that they should switch more often in the treatment condition when they should not switch, and we do not see such an effect in the second column of Table 11. Lastly, a one-tailed t-test also supports that the treatment decreases subjects' probability of switching when they should more than it increases subjects' probability of switching when they should not, i.e., β_1 in Equation (9) is strictly larger than the negative of β_1 in Equation (10) ($p = 0.025$).

6 Concluding Remarks and Implications

In this study, we investigate the role of memory overload in asymmetric under-reaction to belief-challenging versus belief-confirming information. We show that when subjects are exposed to stronger memory overload, they under-react more to belief-challenging information. In contrast, our data does not support that stronger memory overload leads to more under-reaction to belief-confirming information. Moreover, we show that the data is consistent with a mechanism where it is more cognitively taxing to memorize belief-challenging than belief-confirming information.

In addition to the popular view that confirmation bias, or asymmetric updating behavior, is driven by intrinsic preferences for belief-confirming information, our findings demonstrate that the asymmetry also strongly depends on the informational

environment. This lends credence to the growing theoretical literature that details how limited attention and ability could explain a number of behavioral anomalies. In the following, we discuss the implications of our results on economic theory, information campaigns, and managerial decisions.

Modelling Limited Cognitive Resources In this paper, we analyze and test two different cognitive mechanisms in which how memory or cognitive overload affects belief-updating behavior towards belief-confirming and belief-challenging information. Recently, there has been a surge in research on how imperfect cognition influences economic decisions (see the discussion and summary in Enke (2024)). Our results shed light on a novel and understudied characteristic of imperfect cognition: whether cognitive resources are substitutes or independent when processing multiple pieces of information. Our results support the latter characterization but not the former. We view our study as a first step towards answering this question and hope to encourage further theoretical and experimental research to provide additional evidence.

Information Campaigns & Advertisement Information campaigns have become ever so popular because of the advances in technology facilitating the dissemination of information. However, research on information campaigns on different topics such as immigration, health, and energy conservation found mixed results on their effectiveness (Snyder (2007); Espinosa and Stoop (2021); Andor et al. (2022); Facchini et al. (2022)). Our results suggest that one of the reasons for the mixed results is that information campaigns could alter how individuals process belief-confirming versus belief-challenging information. More specifically, merely sending more information without considering individuals' cognitive load disproportionally strengthens under-reaction to belief-challenging information and thus erodes the effectiveness of the information campaigns or even backfires. In contrast, this paper suggests that a better solution is to design information in ways that are less cognitively demanding to process.

Similarly, memory overload could be utilized by incumbent firms as a means to deter entrance. By using excessive advertisement or other forms of information provision, the incumbent strengthens memory overload, such that loyal consumers would under-reaction to information that is in favor of new entrants. We encourage further research on this new role of advertisement.

Managerial Implications Cognitive overload in the workplace has received more attention in recent years. Existing literature shows that cognitive overload leads to an increase in stress and anxiety (da Silva Cezar and Maçada (2023)), and leads to lower productivity (Biondi et al. (2021)). In business, managing cognitive overload has also attracted ample attention: an article on Forbes.com has claimed that

"cognitive overload could be the most important employee experience metric In the next 10 years".[9] This paper shows that memory overload also worsens decision quality, changing decision-making behavior and disproportionally strengthening underreaction to belief-challenging information. Importantly, our treatment v.s. control setting also hints at how to design a workplace to weaken memory overload. More specifically, managers should design information dissemination to reduce cognitive load. For example, instead of presenting multiple pieces of information separately, one should better organize them in a unified framework so workers can easily aggregate the information.

[9]See https://www.forbes.com/councils/forbesbusinessdevelopmentcouncil/2020/06/30/why-cognitive-load-could-be-the-most-important-employee-experience-metric-in-the-next-10-years/.

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A.1 Proofs

A.1.1 Proof of Proposition 1

Proof. We first prove that there exists two thresholds $\overline{s}, \underline{s}$ satisfying $\overline{s} > \underline{s}$ such that the subject memorizes s_1 iff $s_1 \geq \overline{s}$ or $s_1 \leq \underline{s}$.

First, suppose the subject receives $s_1 \geq \frac{1-p_H}{p_H}$, we show that his utility of memorizing minus not memorizing increases in s_1. His utility of memorizing is equal to:

$$\frac{p_H f_H(s_1)}{p_H f_H(s_1) + p_L f_L(s_1)} = \frac{p_H s_1}{p_H s_1 + p_L}.$$

Given the equilibrium probabilty of memorizing in state H and L, denoted as m_H and m_L, the subject's utility of not memorizing is equal to:

$$\frac{p_H s_1}{p_H s_1 + p_L} \left(\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_H(s)\, ds \right) + \frac{p_L}{p_H s_1 + p_L} \left(\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_L(s)\, ds \right)$$

The utility of memorizing s_1 minus not memorizing is equal to:

$$\begin{aligned}
&\frac{p_H s_1}{p_H s_1 + p_L} \left(1 - \int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_H(s)\, ds \right) - \frac{p_L}{p_H s_1 + p_L} \left(\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_L(s)\, ds \right) \\
=&\frac{p_H s_1}{p_H s_1 + p_L} \left(\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_H(s)\, ds \right) - \frac{p_L}{p_H s_1 + p_L} \left(\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_L(s)\, ds \right) \\
=&- \left(\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_L(s)\, ds \right) + \frac{p_H s_1}{p_H s_1 + p_L} \left(\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} (f_H(s) + f_L(s))\, ds \right)
\end{aligned}$$

which increases in s_1 as $\frac{p_H s_1}{p_H s_1 + p_L}$ increases in s_1.

Second, suppose the subject receives $s_1 < \frac{1-p_H}{p_H}$. His utility of memorizing is equal to:

$$\frac{p_L f_L(s_1)}{p_H f_H(s_1) + p_L f_L(s_1)} = \frac{p_L}{p_H s_1 + p_L}.$$

The subject's utility of not memorizing is equal to:

$$\frac{p_H s_1}{p_H s_1 + p_L} \left(\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_H(s)\, ds \right) + \frac{p_L}{p_H s_1 + p_L} \left(\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_L(s)\, ds \right)$$

The utility of memorizing s_1 minus not memorizing is equal to:

$$\begin{aligned}
&-\frac{p_H s_1}{p_H s_1+p_L}\left(\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_H(s)\,ds\right)+\frac{p_L}{p_H s_1+p_L}\left(1-\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_L(s)\,ds\right)\\
=&-\frac{p_H s_1}{p_H s_1+p_L}\left(\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_H(s)\,ds\right)+\frac{p_L}{p_H s_1+p_L}\left(\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_L(s)\,ds\right)\\
=&-\left(\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_H(s)\,ds\right)+\frac{p_L}{p_H s_1+p_L}\left(\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} (f_H(s)+f_L(s))\,ds\right)
\end{aligned}$$

which decreases in s_1 as $\frac{p_L}{p_H s_1+p_L}$ increases in s_1.

Thus, the equilibrium is characterized as two thresholds $\overline{s} \geq \frac{1-p_H}{p_H} > \underline{s}$ which satisfy the following Equations:

$$\begin{aligned}
\overline{s} =&\frac{1-p_H}{p_H}\frac{\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_L(s)\,ds}{\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_H(s)\,ds} \equiv A\left(\frac{1-m_L}{1-m_H}\right)\\
\frac{1}{\underline{s}} =&\frac{p_H}{1-p_H}\frac{\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_H(s)\,ds}{\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_L(s)\,ds} \equiv B\left(\frac{1-m_L}{1-m_H}\right) \qquad (11)\\
1-m_H =&\int_{\underline{s}}^{\overline{s}} f_H(s)\,ds\\
1-m_L =&\int_{\underline{s}}^{\overline{s}} f_L(s)\,ds
\end{aligned}$$

We now show that there always exists an equilibrium where $\overline{s} > \underline{s}$ and $\overline{s} < \frac{1}{\underline{s}}$.

We prove the statement in 4 steps. 1) $\frac{m_L}{m_H}$ decreases and equivalently $\frac{1-m_L}{1-m_H}$ increases when $\overline{s}$ or $\underline{s}$ decreases. Note that

$$\frac{1-m_L}{1-m_H}=\frac{\int_{\underline{s}}^{\overline{s}} f_L(s)\,ds}{\int_{\underline{s}}^{\overline{s}} f_H(s)\,ds}.$$

Note that when $\overline{s}$ decreases, as $\frac{f_L(s)}{f_H(s)}$ decreases in s, $\frac{\int_{\underline{s}}^{\overline{s}} f_L(s)\,ds}{\int_{\underline{s}}^{\overline{s}} f_H(s)\,ds}$ and $\frac{1-m_L}{1-m_H}$ increases. Similarly, when $\overline{s}$ decreases, $\frac{1-m_L}{1-m_H}$ increases.

2) When $\frac{1-m_L}{1-m_H}$ increases, $A(\cdot)$ decreases and $B(\cdot)$ increases. Recall that

$$A\left(\frac{1-m_L}{1-m_H}\right)=\frac{1-p_H}{p_H}\frac{\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_L(s)\,ds}{\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_H(s)\,ds}.$$

Similar to our argument in 1), as $\frac{f_L(s)}{f_H(s)}$ decreases in s, $A\left(\frac{1-m_L}{1-m_H}\right)$ decreases as $\frac{1-m_L}{1-m_H}$

increases. Now, recall that

$$B\left(\frac{1-m_L}{1-m_H}\right) = \frac{p_H}{1-p_H}\frac{\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_H(s)\,ds}{\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_L(s)\,ds}$$

As $\frac{f_H(s)}{f_L(s)}$ increases in s, $B\left(\frac{1-m_L}{1-m_H}\right)$ increases when $\frac{1-m_L}{1-m_H}$ increases.

3) Denote Equation (11) as

$$\begin{pmatrix}\overline{s}\\ \underline{s}\end{pmatrix} = \begin{pmatrix} \frac{1-p_H}{p_H}\frac{\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_L(s)\,ds}{\int_0^{\frac{p_L(1-m_L)}{p_H(1-m_H)}} f_H(s)\,ds} \\ \frac{1-p_H}{p_H}\frac{\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_L(s)\,ds}{\int_{\frac{p_L(1-m_L)}{p_H(1-m_H)}}^{\infty} f_H(s)\,ds} \end{pmatrix} = \tilde{A}\begin{pmatrix}\overline{s}\\ \underline{s}\end{pmatrix}$$

and consider the space where $\overline{s} \in \left[\frac{1-p_H}{p_H}, \frac{1-p_H}{p_H}\frac{\int_0^{\frac{p_L}{p_H}} f_L(s)\,ds}{\int_0^{\frac{p_L}{p_H}} f_H(s)\,ds}\right]$ and $\underline{s} \in \left[0, \frac{1-p_H}{p_H}\frac{\int_{\frac{p_L}{p_H}}^{\infty} f_L(s)\,ds}{\int_{\frac{p_L}{p_H}}^{\infty} f_H(s)\,ds}\right]$.

When $\overline{s} = \frac{1-p_H}{p_H}\frac{\int_0^{\frac{p_L}{p_H}} f_L(s)\,ds}{\int_0^{\frac{p_L}{p_H}} f_H(s)\,ds}$ and $\underline{s} = \frac{1-p_H}{p_H}\frac{\int_{\frac{p_L}{p_H}}^{\infty} f_L(s)\,ds}{\int_{\frac{p_L}{p_H}}^{\infty} f_H(s)\,ds}$, if

$$\frac{1-p_H}{p_H}\frac{\int_0^{\frac{p_L}{p_H}} f_L(s)\,ds}{\int_0^{\frac{p_L}{p_H}} f_H(s)\,ds} < \frac{1}{\frac{1-p_H}{p_H}\frac{\int_{\frac{p_L}{p_H}}^{\infty} f_L(s)\,ds}{\int_{\frac{p_L}{p_H}}^{\infty} f_H(s)\,ds}}$$

we have $1-m_L > 1-m_H$ (by the symmetry of f_H and f_L), and by 1) and 2), $\tilde{A}(\cdot) \leq (\hat{s}, 1/\hat{s})$. And by 1) and 2), when $\overline{s}$ and/or $\underline{s}$ decreases, $\tilde{A}(\cdot)$ decreases in both dimensions. Hence $\tilde{A}$ is a one-to-one mappying from $[\frac{1-p_H}{p_H}, \hat{s}] \times [0, 1/\hat{s}]$ which is a non-empty, compact and convex space. By the Kakutani fixed-point theorem, there exists a fixed point $[\frac{1-p_H}{p_H}, \hat{s}] \times [0, 1/\hat{s}]$, such that $\overline{s} < 1/\underline{s}$.

□

A.1.2 Proof of Proposition 2

Proof. First note that given a signal $s > 1$,

$$\log\frac{p_H^B}{1-p_H^B} - \log\frac{p_H}{1-p_H} = \log s.$$

Given a signal $\tilde{s} = 1/s$,

$$\log\frac{p_H^B}{1-p_H^B} - \log\frac{p_H}{1-p_H} = \log\tilde{s} = -\log s.$$

Now, with the first-order approach, the optimal probability of memorizing a belief-confirming signal $r^*(s)$ for $s > 1$ is given by:

$$r^*(s) = \begin{cases} \frac{(\log s)^2}{2c} & \text{if } (\log s)^2 < 2c; \\ 1 & \text{otherwise.} \end{cases}$$

On the other hand, the optimal probability of memorizing a belief-confirming signal $r^*(\tilde{s})$ for $\tilde{s} = 1/s$ is given by:

$$r^*(\tilde{s}) = \begin{cases} \frac{(\log s)^2}{2\delta c} & \text{if } (\log s)^2 < 2\delta c; \\ 1 & \text{otherwise.} \end{cases}$$

Thus, if $r^*(\tilde{s}) = 1$, $r^*(s) = 1$. If $r^*(s) < 1$, $r^*(s) = \delta r^*(\tilde{s})$. The first part of the Proposition follows.

For the second part of the Proposition, denote the threshold where $r^*(\tilde{s}) = 1$ as $\overline{c}$ and the threshold where $r^*(s) = 1$ as $\underline{c}$. By the first part of the Proposition, we have $\overline{c} > \underline{c}$. We have

$$\frac{\frac{\partial r^*(s)}{\partial c}}{r^*(s)} = \begin{cases} -\frac{(\log s)^2}{2c^2} / -\frac{(\log s)^2}{2c} = -\frac{1}{c} & \text{if } c < \underline{c}; \\ 0 & \text{otherwise.} \end{cases}$$

and,

$$\frac{\frac{\partial r^*(\tilde{s})}{\partial c}}{r^*(\tilde{s})} = \begin{cases} -\frac{(\log s)^2}{2\delta c^2} / -\frac{(\log s)^2}{2\delta c} = -\frac{1}{c} & \text{if } c < \overline{c}; \\ 0 & \text{otherwise.} \end{cases}$$

Thus, when $c \geq \overline{c}$ or $c < \underline{c}$, we have $\frac{\frac{\partial r^*(s)}{\partial c}}{r^*(s)} = \frac{\frac{\partial r^*(\tilde{s})}{\partial c}}{r^*(\tilde{s})}$. When $c \in [\underline{c}, \overline{c})$, we have $\frac{\frac{\partial r^*(s)}{\partial c}}{r^*(s)} = 0 > \frac{\frac{\partial r^*(\tilde{s})}{\partial c}}{r^*(\tilde{s})} = -\frac{1}{c}$. The second part of the Proposition follows. □

Online Appendix of Memory Overload and Asymmetric Under-reaction

Lorenz Goette[1], *Hua-Jing Han*[2], *Benson Tsz Kin Leung*[3]

June 12, 2025

A.2 The distribution in the Guessing Task

We present the reasoning behind the parameters of the distribution. In particular, they are chosen to satisfy the following criteria:

1. The logarithmic odds ratios are monotonic and approximately linear, as shown in Figure A1. That is, higher numbers are stronger evidence that the "high" distribution is true and the differences in the strengths of adjacent signals are approximately constant;

2. After seeing the first sequence of numbers, there are enough subjects with confident belief, i.e., they believe that the state is high (low) with probability 75% or above. Table A.2 shows that more than 40% of the subjects are "confident" after seeing 5 signals. This is to ensure that there exists a significant amount of confident individuals such that confirmation bias could take effect;

3. After seeing the first sequence of numbers, there should not be many subjects with too confident belief, e.g., believe that the state is high(low) with probability 95% or above. Table A.3 shows that less than 2% of the subjects are extremely confident after seeing 5 signals. It ensures that there are sufficient number of observations where a switching occurs, as shown in Table A.1.

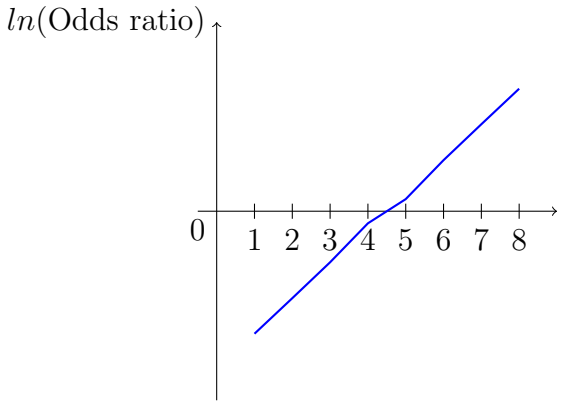


Figure A1: Logarithmic Odds ratios of the numbers 1-8.

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min	mean	max
18.5%	24.2%	29.5%

Table A.1: Simulated proportion of observations where subjects should switch from believing "High" after phase 1 to believing "Low" after phase 2, or from believing "Low" to believing "High", with 10000 simulations of 2000 observations.

		Belief				
		$= 0.5$	> 0.6 or < 0.4	> 0.65 or < 0.35	> 0.7 or < 0.3	> 0.75 or < 0.25
5 draws	Average proportion of subjects	0	76.1%	64.7%	52.5%	41.7%
12 draws	Average proportion of subjects	0%	94.4%	88.8%	76.0%	68.9%

Table A.2: Distribution of Bayesian beliefs given 5 and 12 draws, with 10000 simulations of 2000 observations.

		Belief			
		> 0.8 or < 0.2	> 0.85 or < 0.15	> 0.9 or < 0.1	> 0.95 or < 0.05
5 draws	Average proportion of subjects	32.2%	19.2%	10.1%	1.65%
12 draws	Average proportion of subjects	60.9%	51.35%	40.05%	24.35%

Table A.3: Distribution of Bayesian beliefs given 5 and 12 draws, with 10000 simulations of 2000 observations.

A.3 Analysis with all data, including those with first elicited belief not equaling 0.5

This section presents the analysis with all data, including those with first elicited beliefs not equaling 0.5. Table A.4 and A.5 shows the preliminary analysis of belief after phase 1 and phase 2, respectively. As in the main text, the first two columns of the two tables show that subjects' elicited beliefs after phases 1 and 2 positively and significantly correlate with the Bayesian belief, although they generally under-update with the signals. The last two columns of Table A.5 show that subjects under-react more to belief-challenging information than to belief-confirming information. Lastly, Figure A2a, A2b and A2c shows the distributions of updating mistakes after phase 1 and 2 in the control, and treatment conditions with confirmatory and challenging signals. For updating mistakes after phase 1, a two-sided Kolmogorov–Smirnov test shows that there is no treatment effect ($p = 0.8659$). For updating mistakes after phase 2, a two-sided Kolmogorov–Smirnov test shows that there is indeed a treatment effect: subjects in the treatment condition make larger mistakes than in the control condition with both belief-challenging and belief-confirming information ($p = 0.04767, 0.06989$ respectively).

	(1) Elicited belief	(2) Elicited belief
Bayesian Belief after phase 1	0.774***	0.783***
	(0.016)	(0.018)
Treatment		0.005
		(0.012)
Treatment × Bayesian		-0.019
Belief after phase 1		(0.020)
Age Controls	Yes	Yes
Gender Controls	Yes	Yes
Stat/Econ Controls	Yes	Yes
R^2	0.675	0.675
Observations	2856	2856
Subjects	260	260

Clustered standard errors on subject-level in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

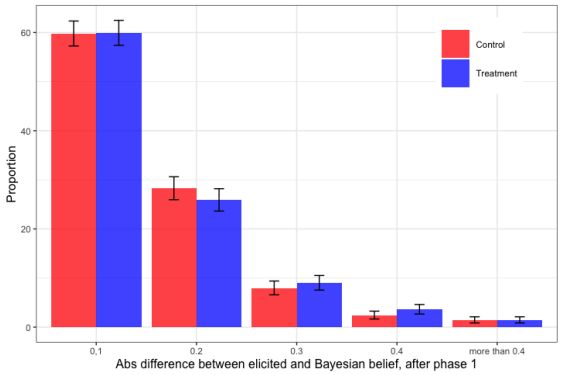
Table A.4: Regression analysis of elicited beliefs after phase 1.Both Bayesian and Elicted beliefs take value of the middle point in their respective 5% bins. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics courses. Subjects in general under-react to information. It corresponds to Table 2.

	(1) Elicited belief	(2) Elicited belief	(3) Elicited belief (w confirmatory signals)	(4) Elicited belief (w challenging signals)
Bayesian Belief after phase 2	0.686***	0.699***	0.754***	0.494***
	(0.013)	(0.015)	(0.014)	(0.036)
Treatment		0.016	0.003	0.048†
		(0.010)	(0.010)	(0.024)
Treatment × Bayesian Belief after phase 2		-0.025	-0.006	-0.082†
		(0.015)	(0.013)	(0.044)
Age Controls	Yes	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes	Yes
R^2	0.698	0.698	0.851	0.287
Observations	2856	2856	1685	1171
Subjects	260	260	260	259

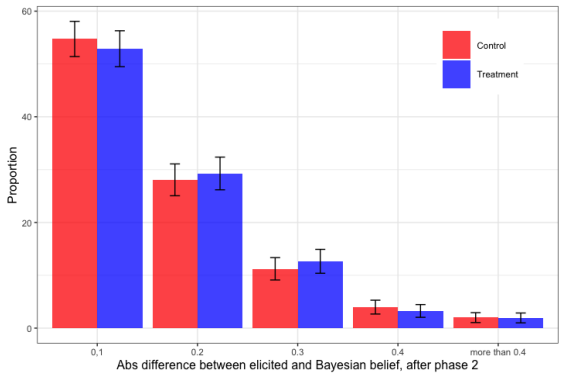
Clustered standard errors on subject-level in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

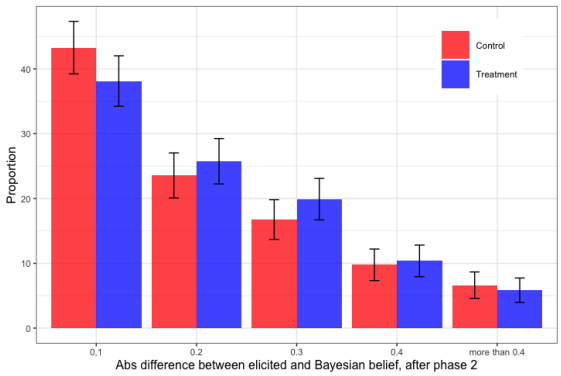
Table A.5: Regression analysis of elicited beliefs after phase 2. Both Bayesian and Elicted beliefs take value of the middle point in their respective 5% bins. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics courses. Similar to phase 1, subjects in general under-react to information. Moreover, they under-react more to challenging signals than to confirmatory signals. It corresponds to Table 3.



(a) Beliefs after Phase 1, treatment versus control.



(b) Beliefs after Phase 2, treatment versus control with confirmatory singals.



(c) Beliefs after Phase 2, treatment versus control with challenging singals.

Figure A2: Distribution of the absolute difference between elicited and Bayesian belief with 95% confidence intervals. Both Bayesian and Elicted beliefs take value of the middle point in their respective 5% bins, and are normalized to values from 0 to 1. Proportions are in percentages. They correspond to Figure 9a, 9b and 9c.

Table A.6, A.7 and A.8 show the OLS analyses on the quantified bias. Similar to the main text, Table A.6 shows that treatment induced a stronger bias in favor of subjects' prior belief than control when they receive belief-challenging information than when they receive belief-confirming information. Different from the main text, the third column of Table A.7 and A.8 show that subjects under-update with both belief-challenging and belief-confirming information in the treatment compared to the control condition, i.e., $p = 0.019$ and $p = 0.011$ respectively. However, comparing the coefficient with belief-challenging information and the negative of the coefficient with belief-confirming signal, it is consistent with our main message: treatment-induced under-reaction to belief-challenging information is larger than the treatment-induced under-reaction to belief-confirming information ($0.166 > -(-0.104)$), although as in the main text, the difference is not significant ($p = 0.216$).

	(1) High prior	(2) Low prior	(3) Pooled
Treatment	-0.099	0.106	-0.099*
	(0.068)	(0.070)	(0.045)
Challenging	1.028***	-1.031***	1.026***
	(0.102)	(0.102)	(0.077)
Treatment × Challenging	0.270*	-0.240	0.263**
	(0.116)	(0.127)	(0.082)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.226	0.230	0.224
Observations	1548	1308	2856
Subjects	259	258	260

Clustered standard errors on subject-level in parentheses

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.6: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics courses. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior data with the quantified bias from the high prior data. This corresponds to Table 4.

Next, Table A.9 and A.10 test the capacity and per-signal cost mechanism, respectively. Similar to the main text, in Table A.9, the coefficient of the interaction term between treatment, challenging dummy for the 1st to 6th signals, and the challenging dummy for the last signal is not significant ($p = 0.910, 0.430, 0.579$ in

	(1) High prior w. confirmatory signals	(2) Low prior w. confirmatory signals	(3) Pooled w. confirmatory signals
Treatment	-0.093	0.134*	-0.104*
	(0.066)	(0.067)	(0.044)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
Pair Controls	Yes	Yes	Yes
R^2	0.718	0.724	0.713
Observations	897	788	1685
Subjects	251	249	260

Clustered standard errors on subject-level in parentheses

† $p < 0.10$,* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.7: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ when subjects receive in aggregate belief-confirming signals in phase 2, i.e., $(p_1 - 0.5) y_{i,obj} < 0$. Standard errors are clustered on subject-level, and we control for subjects' age, gender, whether they have taken Statistics/Economics courses, and include a pairwise fixed effect. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. This corresponds to Table 6.

	(1) High prior w. challenging signals	(2) Low prior w. challenging signals	(3) Pooled w. challenging signals
Treatment	0.154†	-0.189†	0.166*
	(0.091)	(0.107)	(0.064)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
Pair Controls	Yes	Yes	Yes
R^2	0.714	0.737	0.720
Observations	651	520	1171
Subjects	242	233	259

Clustered standard errors on subject-level in parentheses

† $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.8: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ when subjects receive in aggregate belief-challenging signals in phase 2, i.e., $(p_1 - 0.5) y_{i,obj} < 0$. Standard errors are clustered on subject-level, and we control for subjects' age, gender, whether they have taken Statistics/Economics courses, and include a pairwise fixed effect. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. This corresponds to Table 5.

column 1, 2, 3 respectively). Thus, we do not find supportive evidence for the capacity mechanism. On the other hand, in Table A.10, the coefficients of the interaction term between the treatment dummy and the number of phase 2 belief-challenging signals are consistent with the per-signal cost mechanism, and are significant in column 3 where we pool the low prior and high prior data ($p = 0.011, 0.164, 0.005$ in column 1, 2, 3 respectively).

Table A.11 and A.12 present our analysis of the heterogeneous treatment effects. Similar to Table 10, the third column of Table A.12, shows that subjects who are more overwhelmed by our treatment tend to under-react more to belief-confirming signals in treatment vis-a-vis control setting. In contrast to our main text, Table A.11 shows that subjects who are more overwhelmed by our treatment do not under-react more to belief-confirming signals in treatment vis-a-vis control setting. Combined with our findings in our main text, the difference is perhaps due to the confusion or fatigue from subjects who fail the first elicitation test, that they become more conservative and less likely to report more extreme beliefs, especially in the treatment setting.

Lastly, we look at the analysis of subjects' switching decisions. The first column of Table A.13 shows that being in the treatment condition rather than the control condition induces a 9% increase in the probability of not switching when the subject should switch ($p = 0.036$). In contrast, the second column of Table A.13 shows that being in the treatment condition rather than the control condition has no significant impact on the probability of switching when the subject should not switch ($p = 0.434$).

	(1) High prior	(2) Low prior	(3) Pooled
Treatment	-0.182*	0.112	-0.146*
	(0.091)	(0.096)	(0.064)
Challenging 1st-6th	1.126***	-0.865***	1.018***
	(0.121)	(0.115)	(0.089)
Challenging 7th	0.272**	-0.264*	0.268**
	(0.103)	(0.118)	(0.081)
Challenging 1st-6th × Challenging 7th	-0.162	-0.341	0.051
	(0.193)	(0.187)	(0.138)
Treatment × Challenging 1st-6th	0.200	-0.165	0.180
	(0.153)	(0.160)	(0.112)
Treatment × Challenging 7th	0.268	-0.192	0.230*
	(0.141)	(0.151)	(0.108)
Treatment × Challenging 1st-6th × Challenging 7th	-0.026	0.198	-0.096
	(0.227)	(0.251)	(0.172)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.245	0.248	0.238
Observations	1512	1281	2793
Subjects	259	258	260

Clustered standard errors on subject-level in parentheses

† $p < 0.10$,* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.9: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ that analyze whether the treatment effect presents in the last signal. Standard errors are clustered on subject-level. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. This corresponds to Table 7.

	(1) High prior	(2) Low prior	(3) Pooled
Treatment	-0.325*	0.203	-0.265**
	(0.136)	(0.149)	(0.097)
# Challenging	0.352***	-0.323***	0.340***
	(0.034)	(0.037)	(0.027)
Treatment ×	0.104*	-0.064	0.086**
# Challenging	(0.041)	(0.046)	(0.041)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.243	0.204	0.220
Observations	1512	1281	2793
Subjects	259	258	260

Clustered standard errors on subject-level in parentheses

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.10: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ that analyze how the number of challenging signals affects the treatment effect. Standard errors are clustered on subject-level. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. This corresponds to Table 8.

	(1) High prior w. challenging signals	(2) Low prior w. challenging signals	(3) Pooled w. challenging signals
Treatment	0.158	-0.105	0.140*
	(0.090)	(0.106)	(0.063)
Subject-level Mistakes	1.296	-1.968	1.534
(Treatment - Control)	(1.958)	(1.836)	(1.488)
Treatment ×	0.026	-2.029	0.961
Subject-level Mistakes	(2.109)	(2.004)	(1.200)
(Treatment - Control)			
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.032	0.079	0.042
Observations	651	520	1171
Subjects	242	233	259

Clustered standard errors on subject-level in parentheses

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.11: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ on the heterogeneous treatment effect across subjects when they receive in aggregate belief-challenging signals in phase 2. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics Courses. We did not include pairwise fixed effects as they introduce collinearity issues with the subject-level mistakes. "Subject-level mistakes (treatment-control)" is defined as the average mistakes (the absolute difference between elicited and Bayesian belief after phase 2) that subjects make in their six treatment rounds minus that in their six control rounds. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. This corresponds to Table 9.

	(1) High prior w. confirmatory signals	(2) Low prior w. confirmatory signals	(3) Pooled w. confirmatory signals
Treatment	-0.096	0.072	-0.078
	(0.073)	(0.070)	(0.046)
Subject-level Mistakes	1.058	-3.854***	2.434**
(Treatment - Control)	(0.960)	(1.020)	(0.759)
Treatment ×	-1.079	3.725**	-2.395***
Subject-level Mistakes	(1.148)	(1.156)	(0.660)
(Treatment - Control)			
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.035	0.080	0.037
Observations	897	788	1685
Subjects	251	249	260

Clustered standard errors on subject-level in parentheses

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.12: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ on the heterogeneous treatment effect across subjects when they receive in aggregate belief-confirming signals in phase 2. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics Courses. We did not include pairwise fixed effects as they introduce collinearity issues with the subject-level mistakes. "Subject-level mistakes (treatment-control)" is defined as the average mistakes (the absolute difference between elicited and Bayesian belief after phase 2) that subjects make in their six treatment rounds minus that in their six control rounds. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. This corresponds to Table 10.

	(1) Should switch but didn't	(2) Shouldn't switch but did
Treatment	0.090*	0.008
	(0.042)	(0.010)
Age Controls	Yes	Yes
Gender Controls	Yes	Yes
Stat/Econ Controls	Yes	Yes
Group Controls	Yes	Yes
R^2	0.773	0.689
Observations	731	2125
Subjects	244	260

Clustered standard errors on subject-level in parentheses

Pairwise fixed effects

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.13: Regression analysis of Switching Decisions after phase 2. Standard errors are clustered on subject-level, and we control for subjects' age, gender, whether they have taken Statistics/Economics courses, and include pairwise fixed effects. The first column focuses on cases where a Bayesian should switch from guessing high in phase 1 to low in phase 2 or guessing low in phase 1 to high in phase 2, but the subject did not switch. The second column focuses on cases where a Bayesian should not switch, but the subject has switched from guessing high to low or guessing low to high. The dependent variables are the dummy variables that denote such switching mistakes. This corresponds to Table 11.

A.4 Analysis with data excluding all observations in the first 4 sessions

This section presents the analysis excluding all data from sessions 1 to 4, affected by the technological glitch. Table A.14 and A.15 shows the preliminary analysis of belief after phase 1 and phase 2, respectively. As in the main text, the first two columns of the two tables show that subjects' elicited beliefs after phases 1 and 2 positively and significantly correlate with the Bayesian belief, although they generally under-update with the signals. The last two columns of Table A.15 show that subjects under-react more to belief-challenging information than to belief-confirming information. Lastly, Figure A3a, A3b and A3c shows the distributions of updating mistakes after phase 1 and 2 in the control, and treatment conditions with confirmatory and challenging signals. For updating mistakes after phase 1, a two-sided Kolmogorov–Smirnov test shows that there is no treatment effect ($p = 0.963$). For updating mistakes after phase 2, a two-sided Kolmogorov–Smirnov test shows that there is indeed a treatment effect: subjects in the treatment condition make larger mistakes than in the control condition with both belief-challenging and belief-confirming information ($p = 0.144, 0.295$ respectively).

	(1) Elicited belief	(2) Elicited belief
Bayesian Belief after phase 1	0.764***	0.779***
	(0.019)	(0.022)
Treatment		0.016
		(0.015)
Treatment × Bayesian		-0.031
Belief after phase 1		(0.025)
Age Controls	Yes	Yes
Gender Controls	Yes	Yes
Stat/Econ Controls	Yes	Yes
R^2	0.699	0.700
Observations	1646	1646
Subjects	160	160

Clustered standard errors on subject-level in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

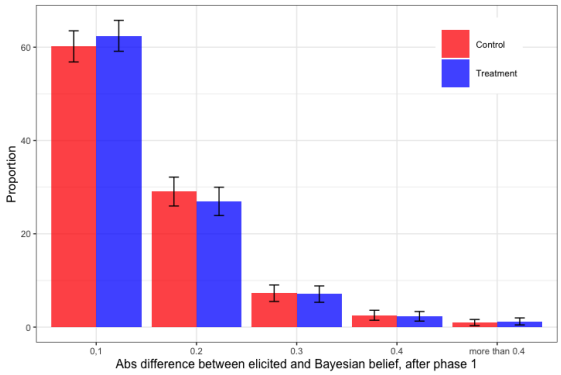
Table A.14: Regression analysis of elicited beliefs after phase 1.Both Bayesian and Elicted beliefs take value of the middle point in their respective 5% bins. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics courses. Subjects in general under-react to information. It corresponds to Table 2.

	(1) Elicited belief	(2) Elicited belief	(3) Elicited belief (w confirmatory signals)	(4) Elicited belief (w challenging signals)
Bayesian Belief after phase 2	0.682***	0.694***	0.753***	0.499***
	(0.016)	(0.020)	(0.018)	(0.037)
Treatment		0.014	-0.001	0.059*
		(0.013)	(0.014)	(0.027)
Treatment × Bayesian Belief after phase 2		-0.025	-0.003	-0.111*
		(0.018)	(0.017)	(0.047)
Age Controls	Yes	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes	Yes
R^2	0.716	0.716	0.856	0.336
Observations	1646	1646	967	679
Subjects	160	160	157	153

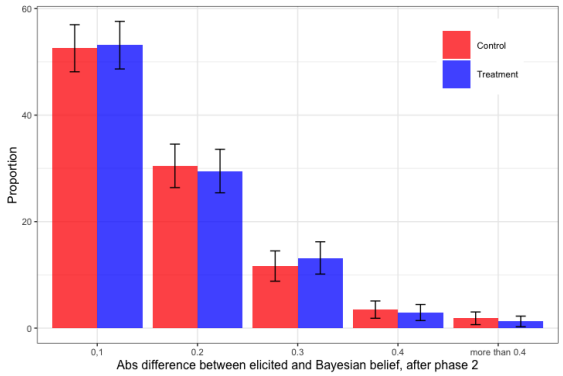
Clustered standard errors on subject-level in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

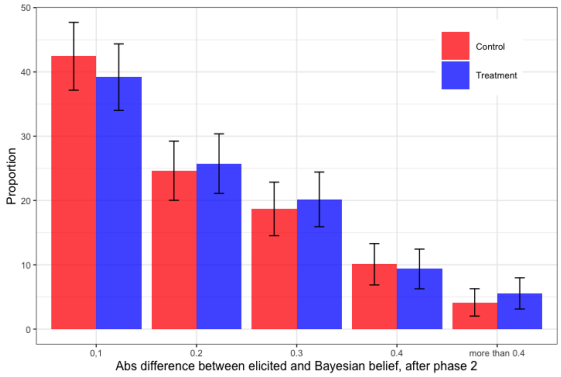
Table A.15: Regression analysis of elicited beliefs after phase 2. Both Bayesian and Elicted beliefs take value of the middle point in their respective 5% bins. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics courses. Similar to phase 1, subjects in general under-react to information. Moreover, they under-react more to challenging signals than to confirmatory signals. It corresponds to Table 3.



(a) Beliefs after Phase 1, treatment versus control.



(b) Beliefs after Phase 2, treatment versus control with confirmatory singals.



(c) Beliefs after Phase 2, treatment versus control with challenging singals.

Figure A3: Distribution of the absolute difference between elicited and Bayesian belief with 95% confidence intervals. Both Bayesian and Elicted beliefs take value of the middle point in their respective 5% bins, and are normalized to values from 0 to 1. Proportions are in percentages. They correspond to Figure 9a, 9b and 9c.

Table A.16, A.17 and A.18 show the OLS analyses on the quantified bias. Similar to the main text, Table A.16 shows that treatment induced a stronger bias in favor of subjects' prior belief than control when they receive belief-challenging information than when they receive belief-confirming information. Different from the main text, the third column of Table A.17 and A.18 show that subjects under-update with belief-challenging but not with belief-confirming information in the treatment compared to the control condition, i.e., $p = 0.021$ and $p = 0.83$ respectively. Comparing the coefficients with belief-challenging information and the negative of the coefficient with belief-confirming signal, it is consistent with our main message: treatment-induced under-reaction to belief-challenging information is larger than the treatment-induced under-reaction to belief-confirming information ($0.166 > -(-0.015)$) which is supported by a one-tailed t-test ($p = 0.043$).

	(1) High prior	(2) Low prior	(3) Pooled
Treatment	-0.108	0.050	-0.066
	(0.083)	(0.082)	(0.055)
Challenging	1.161***	-1.161***	1.163***
	(0.127)	(0.102)	(0.095)
Treatment × Challenging	0.237	-0.214	0.223*
	(0.147)	(0.136)	(0.102)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.227	0.293	0.276
Observations	909	737	1646
Subjects	157	151	160

Clustered standard errors on subject-level in parentheses
† $p < 0.10$,* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.16: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics courses. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior data with the quantified bias from the high prior data. This corresponds to Table 4.

Next, Table A.19 and A.20 test the capacity and per-signal cost mechanism, respectively. Similar to the main text, in Table A.19, the coefficient of the interaction term between treatment, challenging dummy for the 1st to 6th signals, and the challenging dummy for the last signal, is not significant ($p = 0.114, 0.480, 0.596$ in column 1, 2, 3 respectively). Thus, we do not find supportive evidence for the

	(1) High prior w. confirmatory signals	(2) Low prior w. confirmatory signals	(3) Pooled w. confirmatory signals
Treatment	-0.122	-0.033	-0.015
	(0.103)	(0.094)	(0.068)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
Pair Controls	Yes	Yes	Yes
R^2	0.801	0.823	0.799
Observations	530	437	967
Subjects	149	143	157

Clustered standard errors on subject-level in parentheses

† $p < 0.10$,* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.17: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ when subjects receive in aggregate belief-confirming signals in phase 2, i.e., $(p_1 - 0.5)y_{i,obj} < 0$. Standard errors are clustered on subject-level, and we control for subjects' age, gender, whether they have taken Statistics/Economics courses, and include a pairwise fixed effect. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. This corresponds to Table 6.

	(1) High prior w. challenging signals	(2) Low prior w. challenging signals	(3) Pooled w. challenging signals
Treatment	0.172†	-0.189†	0.203*
	(0.099)	(0.107)	(0.086)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
Pair Controls	Yes	Yes	Yes
R^2	0.714	0.737	0.720
Observations	379	520	679
Subjects	142	233	153

Clustered standard errors on subject-level in parentheses

† $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.18: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ when subjects receive in aggregate belief-challenging signals in phase 2, i.e., $(p_1 - 0.5)y_{i,obj} < 0$. Standard errors are clustered on subject-level, and we control for subjects' age, gender, whether they have taken Statistics/Economics courses, and include a pairwise fixed effect. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases.This corresponds to Table 5.

capacity mechanism. On the other hand, in Table A.20, the coefficients of the interaction term between the treatment dummy and the number of phase 2 belief-challenging signals are consistent with the per-signal cost mechanism and the p-value in column 3 is only slightly higher than the 5% significance level when we pool the low prior and high prior data ($p = 0.125, 0.146, 0.051$ in column 1, 2, 3 respectively).

Table A.11 and A.12 present our analysis of the heterogeneous treatment effects. Similar to Table 10, the third columns of Table A.11 and Table A.12 show that subjects who are more overwhelmed by our treatment tend to under-react more to both belief-challening informatio and belief-confirming signals in treatment vis-a-vis control setting. Similar to the main text, when comparing the coefficient β_3 with belief-challenging signals and the negative of β_3 with belief-confirming signals, the magnitude of under-reaction induced by memory overload (treatment) and subject-level mistakes is larger with belief-challenging signals than with belief-confirming signals (e.g., $3.802 > -(-2.652)$ in column 3), albeit the difference is not significant ($p = 0.553$ in a two-sided t-test).

Lastly, we look at the analysis of subjects' switching decisions. The first column of Table A.23 shows that being in the treatment condition rather than the control condition induces an 11.3% increase in the probability of not switching when the subject should switch (although $p = 0.070$ is slightly higher than the 5% significance level). In contrast, the second column of Table A.23 shows that being in the treatment condition rather than the control condition has no significant impact on the probability of switching when the subject should not switch ($p = 0.386$).

	(1) High prior	(2) Low prior	(3) Pooled
Treatment	-0.102	0.065	-0.081
	(0.116)	(0.114)	(0.083)
Challenging 1st-6th	1.421***	-0.954***	1.223***
	(0.153)	(0.144)	(0.118)
Challenging 7th	0.451**	-0.207	0.333**
	(0.135)	(0.149)	(0.102)
Challenging 1st-6th × Challenging 7th	-0.550*	-0.384	-0.131
	(0.238)	(0.214)	(0.162)
Treatment × Challenging 1st-6th	0.015	-0.080	0.063
	(0.188)	(0.215)	(0.150)
Treatment × Challenging 7th	0.043	-0.291	0.158
	(0.187)	(0.193)	(0.136)
Treatment × Challenging 1st-6th × Challenging 7th	0.445	0.223	0.107
	(0.280)	(0.314)	(0.201)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.307	0.295	0.287
Observations	883	718	1601
Subjects	156	151	160

Clustered standard errors on subject-level in parentheses

† $p < 0.10$,* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.19: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ that test the prediction of the capacity mechanisms. Standard errors are clustered on subject-level, and we control for subjects' age, gender, whether they have taken Statistics/Economics courses. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. "Challenging 1st-6th" is a dummy variable that equals to 1 if the 6 signals are in aggregate belief-challenging. "Challenging 7th" is a dummy variable that equals to 1 if the 7th signal are in aggregate belief-challenging. This corresponds to Table 7.

	(1) High prior	(2) Low prior	(3) Pooled
Treatment	-0.269	0.172	-0.205
	(0.176)	(0.169)	(0.121)
# Challenging	0.395***	-0.360***	0.384***
	(0.042)	(0.046)	(0.035)
Treatment ×	0.082	-0.075	0.076†
# Challenging	(0.053)	(0.051)	(0.039)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.305	0.268	0.282
Observations	883	718	1601
Subjects	156	151	160

Clustered standard errors on subject-level in parentheses

† $p < 0.10$,* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.20: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ that analyze how the number of challenging signals affects the treatment effect. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics Courses. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. # Challenging$_i$ is the number of challenging signals subject receives in phase 2. This corresponds to Table 8.

	(1) High prior w. challenging signals	(2) Low prior w. challenging signals	(3) Pooled w. challenging signals
Treatment	0.103	-0.118	0.119
	(0.104)	(0.116)	(0.078)
Subject-level Mistakes (Treatment - Control)	-0.068	0.842	-0.743
	(3.758)	(1.653)	(2.577)
Treatment × Subject-level Mistakes (Treatment - Control)	0.771	-6.370***	3.802†
	(3.834)	(1.664)	(1.940)
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.045	0.135	0.059
Observations	379	300	679
Subjects	142	134	153

Clustered standard errors on subject-level in parentheses

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.21: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ on the heterogeneous treatment effect across subjects when they receive in aggregate belief-challenging signals in phase 2. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics Courses. We did not include pairwise fixed effects as they introduce collinearity issues with the subject-level mistakes. "Subject-level mistakes (treatment-control)" is defined as the average mistakes (the absolute difference between elicited and Bayesian belief after phase 2) that subjects make in their six treatment rounds minus that in their six control rounds. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. This corresponds to Table 9.

	(1) High prior w. confirmatory signals	(2) Low prior w. confirmatory signals	(3) Pooled w. confirmatory signals
Treatment	-0.118	0.014	-0.043
	(0.090)	(0.080)	(0.058)
Subject-level Mistakes	0.532	-5.645***	3.050**
(Treatment - Control)	(1.219)	(1.181)	(1.042)
Treatment ×	-0.015	5.075***	-2.652**
Subject-level Mistakes	(1.355)	(1.308)	(0.901)
(Treatment - Control)			
Age Controls	Yes	Yes	Yes
Gender Controls	Yes	Yes	Yes
Stat/Econ Controls	Yes	Yes	Yes
R^2	0.041	0.146	0.046
Observations	530	437	967
Subjects	149	143	157

Clustered standard errors on subject-level in parentheses

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.22: Regression analysis of the quantified bias $\log y_{i,sub} - \log y_{i,obj} = \log \frac{p_2^B}{1-p_2^B} - \log \frac{p_2}{1-p_2}$ on the heterogeneous treatment effect across subjects when they receive in aggregate belief-confirming signals in phase 2. Standard errors are clustered on subject-level, and we control for subjects' age, gender and whether they have taken Statistics/Economics Courses. We did not include pairwise fixed effects as they introduce collinearity issues with the subject-level mistakes. "Subject-level mistakes (treatment-control)" is defined as the average mistakes (the absolute difference between elicited and Bayesian belief after phase 2) that subjects make in their six treatment rounds minus that in their six control rounds. The first column focuses on the data where subjects have a high prior $p_1 \geq 0.5$; the second column focuses on the data where subjects have a low prior $p_1 < 0.5$; the third column pooled the negative of the quantified bias from the low prior cases with the positive of the quantified bias from the high prior cases. This corresponds to Table 10.

	(1) Should switch but didn't	(2) Shouldn't switch but did
Treatment	0.113†	-0.012
	(0.061)	(0.014)
Age Controls	Yes	Yes
Gender Controls	Yes	Yes
Stat/Econ Controls	Yes	Yes
Group Controls	Yes	Yes
R^2	0.852	0.779
Observations	452	1194
Subjects	141	159

Clustered standard errors on subject-level in parentheses

Pairwise fixed effects

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.23: Regression analysis of Switching Decisions after phase 2. Standard errors are clustered on subject-level, and we control for subjects' age, gender, whether they have taken Statistics/Economics courses, and include pairwise fixed effects. The first column focuses on cases where a Bayesian should switch from guessing high in phase 1 to low in phase 2 or guessing low in phase 1 to high in phase 2, but the subject did not switch. The second column focuses on cases where a Bayesian should not switch, but the subject has switched from guessing high to low or guessing low to high. The dependent variables are the dummy variables that denote such switching mistakes. This corresponds to Table 11.